Retail Trading and Return Predictability in China

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Abstract

Using comprehensive account-level data, we separate Chinese retail investors into five groups and document strong heterogeneity in trading dynamics and performances. Retail investors with smaller account sizes cannot predict future returns correctly, display daily momentum patterns, fail to process public news, and show overconfidence and gambling preferences; while retail investors with larger account balances predict future returns correctly, display contrarian patterns, and incorporate public news in trading. With Barber et al. (2009) performance measures, smaller retail investors suffer from poor stock selection abilities and trading costs, while large retail investors' stock selection abilities are offset by trading costs.

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Retail investors are important participants in financial markets, and many studies using data from the United States (U.S.) and Europe are devoted to understanding their trading motives, performance, and role in information transmission and price discovery.[1] China's equity market, which is the second largest in the world, provides an important setting for studying retail investors. According to the annual report of the Shanghai Stock Exchange in 2016, retail investors contribute about 85% of the daily trading volume on the exchange, whereas institutional investors contribute only around 15%. Behind the high trading volumes are tens of millions of retail investors in China, which accounts for the largest population of retail investors in the world. The dominance and prevalence of retail trading in China brings retail investors to the central stage of the capital market, making it crucial for researchers, regulators, and practitioners to understand their investment choices and how those choices affect information transmission and price discovery.

An easy reference for answering these questions might appear to be the earlier studies that use data from the U.S. or Europe. However, the development of the Chinese stock market has been substantially different from that of the U.S. or Europe. For instance, the U.S. market displays typical features of developed capital markets, such as the dominance of institutional investors in terms of trading and holding, mature information environment, and established legislature system. Meanwhile, Chinese stock market displays typical features of emerging markets, with rising institutional investors accounting for only a small part of the daily trading volume, high valuations (or lower earnings yields), high volatilities, and high turnovers. Given these significant differences, it is not clear whether previous studies conclusions regarding retail investors in other developed

[1] Related studies include Barber and Odean (2000, 2001, 2008), Barber, Odean and Zhu (2008), Kaniel, Saar, and Titman (2008), Kelley and Tetlock (2013), Welch (2022), Barber, Huang, Odean and Schwartz (2022), Boehmer, Jones, Zhang, and Zhang (2021), Eaton, Green, Roseman and Wu (2022), and many others.

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countries would readily apply to China, and a focused study of Chinese retail investors become essential.

Our study focuses on two core questions to understand retail investors in China: First, how do retail investors contribute to the price discovery process? In other words, can they predict or fail to predict future price movements? Second, if retail investors succeed or fail in predicting future price movements, what are the driving forces of their trading behaviors? Liquidity, information, or behavioral biases? Clearly, millions of Chinese retail investors in our sample are not homogenous individuals, and the answers to these two questions likely differ drastically for different groups of retail investors. Therefore, we rely on the rich cross section of our data to examine the heterogeneity of retail investors and how their trading interacts with stock returns, information flows, and behavioral features.

We are grateful that a major stock exchange in China provides us the access to proprietary account-level trading and holdings data from 2016 to 2019, accounting for over 53 million retail accounts. To comply with regulatory requirements, the exchange categorizes all Chinese retail accounts into five groups based on their account balances: less than 100,000 CNY (RT1), between 100,000 and 500,000 CNY (RT2), between 500,000 and 3,000,000 CNY (RT3), between 3,000,000 and 10,000,000 CNY (RT4), and greater than 10,000,000 CNY (RT5). The five groups account for 58.7%, 28.6%, 10.9%, 1.4%, and 0.4% of the total number of accounts, respectively. In other words, the majority of Chinese retail investors have account sizes below 500,000 CNY. [2]

To answer the first research question on how retail investors contribute to price discovery, we directly examine whether retail investors' buy and sell activities predict future stock returns. If the market is perfectly efficient, stock prices would follow random walks, and retail trading would

[2] Additional gender and age information shows that most Chinese retail investors are young or middle-aged males.

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not predict future returns. If the market is not perfectly efficient and some investors have value-relevant information for future stock prices, their order flows would positively predict future returns. On the other hand, if some investors have an information disadvantage, or fail to incorporate timely information into their trades, their order flows might negatively predict future returns. Using daily retail order imbalances from each retail group, we examine their predictive power for future stock returns at different horizons. The smaller retail investors, RT1-RT4, predict next-day returns with negative and significant coefficients. That is, the prices of stocks they purchase experiences negative returns the following day, whereas the stocks they sell experiences positive returns. By contrast, the largest retail investors, RT5, positively and significantly predict next-day returns, indicating that they purchase and sell stocks in directions consistent with future price movements. When we examine longer horizons, the above-mentioned predictive patterns persist for about 8 to 9 weeks. These patterns remain quite robust when we form long-short strategies on order flow information, and for subsets of stocks with differences in size, value, liquidity, and share price levels.

The data allow us to examine the return predictive patterns among the characteristics that are not easily available in most other datasets. For instance, we directly observe the counterparties of each trade and find that the order flows from small retail investors always predict returns negatively and do not depend on counterparties. Interestingly, for large retail and institutional investors, their order flows typically predict future returns positively, but they become negative when they trade on the same side as small retail investors, suggesting that their prediction becomes erroneous when on the same side as smaller retail investors. We also examine how holding horizons are related to the predictive patterns of retail order flow. Small retail investors' negative return predictive power is stronger over the short horizon and slowly diminishes when holding

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horizons become longer, indicating that a longer holding horizon might lead to better returns. In contrast, large retail investors' positive return predictive power is stronger over the short horizon, which suggests that they might trade on time-sensitive information, and the predictive power becomes weaker for longer horizons.

For the second research question regarding trading motives of retail investors, previous literature provides multiple explanations, such as order flow persistence, liquidity provision, behavioral biases, and information (dis)advantages. These explanations also naturally connect with the predictive pattern of retail order flows for future returns. We adopt the two-stage decomposition procedure in Boehmer et al. (2021) to examine whether these hypotheses explain the trading activities of different retail investor groups and how these trading activities contribute to the predictive patterns for future returns. Our results show that small retail investors display significant order flow persistence, and that their trades have momentum patterns at daily horizons that demand immediate liquidity. Smaller retail investors also display strong behavioral biases, such as overconfidence and gambling preferences, and fail to predict and process informational earnings news. In explaining order flow's predictive power for future returns, order persistence, daily momentum trading patterns, behavioral biases, and information disadvantages all contribute to the negative predictive power of smaller retail investors. In contrast, the largest retail investors display contrarian trading patterns, trade against the behavioral biases of other retail groups, and are capable of predicting and processing earnings news. All of these features contribute to the positive predictive power of the largest retail investors.

To further understand the performance of retail trading flows, we follow Barber et al. (2009) to construct net buy and sell portfolios, track the net positions of each group of retail investors, and compute the performance of the net buy and sell portfolios. Consistent with previous findings

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that smaller retail investors negatively predict future returns, while large investors positively predict future returns, we find that the net buy and sell portfolios tracking smaller retail investors' order flows have an annual return of -5.61%, while the net buy and sell portfolios tracking large retail investors have an annual return of -0.29%. We also decompose the performance into three components: stock selection, market timing, and trading cost. We find that stock selection and trading costs are the two major drivers of total returns in the tracking portfolios.

This study is closely related to the retail investor literature. As mentioned earlier, previous studies on retail investors primarily use data from the U.S. and other markets, and tend to treat retail investors as one group.[3] For instance, using data from a discount broker in the U.S., Barber and Odean (2000, 2001, 2008) document many behavioral biases. Kaniel et al. (2008), Barber et al. (2009), Kelley and Tetlock (2013), and Boehmer et al. (2021) use different datasets from the U.S. and find that retail trading can positively predict the cross-section of future returns. Outside the U.S., Grinblatt and Keloharju (2000), Linnainmaa (2010), and Grinblatt, Keloharju, and Linnainmaa (2012) focus on Finland data; Bach, Calvet, and Sodini (2020) focus on Swedish data; Dorn, Huberman, and Sengmueller (2008) study data from Germany. All of these studies provide important results regarding retail investors' trading activities. Interestingly though, the results are not always consistent, and evidence from China will provide more insights into this literature.

This study is also related to the recent studies of the rapidly growing Chinese stock market. Liu, Stambaugh and Yuan (2019), and Liu, Zhou, and Zhu (2021) establish asset pricing factors for stock returns. For Chinese retail investors, An, Lou, and Shi (2022) study the wealth redistribution role of financial bubbles and crashes over July 2014 and December 2015,

[3] To facilitate reading, Appendix Table I provides a summary table of previous papers on retail investors using data from different markets, and Appendix Table II provides comparisons between the findings of this study with the results from previous literature.

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documenting a net transfer of 250 billion CNY from the poor to ultra-wealthy retail investors over this period. Liu, Peng, Xiong, and Xiong (2021) and Liao, Peng, and Zhu (2021) both focus on the behavioral features of Chinese retail investors and document overconfidence, gambling preferences, and extrapolative expectations in these investors. Li, Geng, Subrahmanyam, and Yu (2017), Chen, Gao, He, Jiang, and Xiong (2019), Jiang, Liu, Peng, and Wang (2022), Hu, Liu, and Xu (2021), and Titman, Wei, and Zhao (2022) all focus on an earlier Chinese sample period and examine behavioral biases and reactions to corporate events. These Chinese studies primarily rely on low-frequency data, or data from a small number of brokerages covering a small part of the market, or investigate issues other than return predictability. As a result, there remains no direct study on the heterogeneity of retail investors' trading behavior, their return predictive power, and how they process information using high-frequency trading data in one major stock market dominated by retail investors.

Compared with previous studies, this study makes three important contributions. First, our study, with its large market coverage of the market for a recent sample period, is one of the most thorough and comprehensive studies of Chinese retail investors, and provides many important implications for regulators, practitioners, and academic researchers. Second, we separate retail investors into groups based on account size and provide unique and direct evidence of investor heterogeneity in terms of return predictability, counterparties, holding horizons, and performance. Third, we examine different hypotheses regarding return prediction patterns for different retail investor groups and provide clear evidence on the sources of the negative or positive predictive power of different retail investors.

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I. Data

A. Data on Retail Investors

Our data comes from one major stock exchange in China, which contains the trading and holding history of all stocks listed on the exchange from all investors, between January 2016 and June 2019. This proprietary dataset contains roughly 53 million accounts, and based on investor identities, they are grouped into three major categories: retail (RT), institutional (INST), and corporations (CORP). Retail investors are further stratified into five groups based on their account sizes at the beginning of each year, which is the average portfolio value (including equity holdings in all A-share listed firms, plus cash balance) over the previous twelve months.[4] As mentioned in the introduction, there are five subgroups: below 100,000 CNY (RT1), 100,000-500,000 CNY (RT2), 500,000 - 3 million CNY (RT3), 3 million - 10 million CNY (RT4), and above 10 million CNY (RT5). Since the focus of this study is how retail trades are related to stock prices in the cross-section, we sum up individual investors' trading information at seven investor group levels (RT1-RT5, INST, and CORP) for each stock each day.

Panel A of Table I presents the summary statistics for investor accounts. During the sample period, the total number of active accounts for retail investors, institutions, and corporations are 53.4 million, 40,000, and 47,000, respectively. Within the retail investor category, there are 31.4 million, 15.3 million, 5.8 million, 0.7 million, and 0.2 million accounts for RT1 to RT5, respectively. Clearly, most of the retail investors have accounts less than 500,000 CNY. The overall trading volume on this exchange averages 201 billion CNY per day, with retail investors,

[4] The annual calculation of account balances is designed by the exchange. There is a concern that investors might migrate to different groups after the initial grouping at the beginning of the year. We acknowledge this possibility. However, the fact that the grouping is redone each year eases the concern to some extent. The cutoff numbers to separate retail investors into different groups are also chosen by the exchange to comply with exchange regulations. For instance, derivatives and leverage trading are only allowed for investors with account size higher than 0.5 million CNY in order to protect the investors with lower account balances.

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institutions, and corporations accounting for 81%, 17%, and 2% of the total trading volume, respectively. Within the retail investor sector, the trading volumes for RT1 to RT5 are 5%, 17%, 27%, 13%, and 19% of the total trading volume, respectively, indicating that RT3 is the most active trading subgroup. Regarding stock holdings, retail investors' holdings account for 22%, institutions are 17%, and corporations are 62%. Within the retail investor sector, the account values for RT1 to RT5 are 1%, 4%, 6%, 3%, and 7% of the total tradable market cap, respectively. With low holdings and high trading volumes, Chinese retail investors trade quite actively; in some cases, they might even trade excessively.

[Insert Table I about here]

To understand the relative importance of different investment groups' trading over time, we plot the time series of cross-sectional means of various investors' trading activities in Figure I. Panel A presents each group's trading volume as a percentage of the total trading volume. Again, the RT3 group has the highest trading volume, accounting for about 30% of total trading. Interestingly, institutional trading gradually increases over time, from 10% in 2016 to over 20% in 2019. The corporations barely trade and account for a negligible amount of trading volume. Panel B displays the percentage of shares held by each group, and it can be seen that the time-series patterns in holdings are stable. Corporations hold a large chunk of shares, accounting for over 60% of total shares; institutions hold about 17% shares; large and medium retail investors hold about 3% to 6% shares, and the smallest retail investors RT1 hold around 1% of the shares. In Panel C, we compute relative trading activeness with respect to holdings for each group, by scaling each group's trading share volumes by each group's holdings at stock level. The retail investors all follow similar time-series patterns, trading around 10% of their holdings each day on average, which is quite high. We observe a slow decrease in trading after 2018 to about 5%, but there is also

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a large but short-term surge up to 15% at the beginning of 2019. Considering that the market return is -22.2% during 2018, and it reverts back to 21.2% during the first half of 2019, it is likely that the large negative returns reduce retail trading enthusiasms, while large positive returns reignite the enthusiasms. By the end of the 2019, retail trading returns to the average level of 10% of holdings. We notice that among the five groups of retail investors, RT5 trade slightly less than the rest of the retail investors. We also observe a clear increasing pattern in the activeness of institutional trading, which increases from 7% in 2016 to 16% in 2019. Finally, the corporates in general rarely trade, and their trading accounts for about 2% of their holdings.

Overall, in the Chinese stock market, retail investors dominate in trading, while corporations dominate in holdings. The retail investors' dominance in trading in the Chinese stock market is likely the joint result of market development, regulations, and investor preferences. This pattern is not particularly rare for emerging markets, but is quite different from that of developed markets.[5]

[Insert Figure I about here]

B. Data on Stock Returns and Firm Characteristics

We obtain data on stock returns, volumes, and accounting information from Wind Information Inc. (WIND), the largest financial data provider in China. For consistency with the retail data, the sample period runs from January 2016 to June 2019. We adopt the filters in Liu et al. (2019) and exclude stocks with less than 15 days of trading records during the most recent

[5] We present additional summary statistics on retail trading volumes and holdings in Appendix Table III. The results show that small retail investors prefer to trade and hold small, low earning/price ratio, and high turnover firms, while the largest retail investors' trading and holdings are tilted towards larger, high earnings-to-price ratio firms. In terms of sectors, the small retail investors prefer to trade and hold the alternative energy sector, and prefer not to trade and hold banks and life insurance firms, while the institutions and corporates behave in opposite ways. Finally, small retail investors tend to use small order sizes, while large retail investors tend to use large orders sizes, while institutions use all order sizes.

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month. Liu et al. (2019) also eliminate stocks that become public within the past six months, stocks with fewer than 120 days of trading records during the past 12 months, and the smallest 30% of all firms listed in the Chinese A-share market. This study does not exclude these stocks from the main results because retail investors trade actively in small stocks and during a stock's IPO period. We present the results with all the filters from Liu et al. (2019) in our robustness checks, and the findings are similar with the additional filters. Starting from March 31, 2010, margin buy and short-sell are allowed on Chinese stock exchanges for subsets of stocks. We include these leveraged trades in our main results and provide an additional analysis that excludes leveraged trading in our robustness checks. We merge the exchange and WIND data by stock ticker and present account summary statistics. Our final sample covers over 1.1 million stock-day observations, and on each day, we have an average of around 1,200 firms.

Table I Panel B presents several key variables for the Chinese capital market. Market capitalization is the product of the previous month's closing price and total A shares outstanding. The average Chinese firm capitalization is 20.1 billion CNY, or 3 billion USD, about half of the cross-sectional average in the U.S stock market during the same period, which is 6.9 billion USD. The earnings-to-price ratio (EP ratio) is the ratio of the most recently reported quarterly net profit, excluding non-recurrent gains/losses over last month-end's market capitalization. According to Liu et al. (2019), the EP ratio captures the value effect. The average EP ratio is 0.0075 in China, while the average EP ratio is 0.0272 in U.S stock market, which indicates high valuation ratios in China. The average daily stock return is -0.01% for Chinese stocks, while the average daily stock return is 0.04% in the U.S. stock market over the same sample period.[6] Finally, we compute daily

[6] Previous literature using the U.S. data shows that microstructure frictions can generate noise in daily return measures. For instance, Blume and Stambaugh (1983) show that daily returns computed from the end-of-day closing prices can have an upward bias due to bid-ask bounce. We compute this bias measure using the closing bid and ask prices for all A-share stocks listed on this major stock exchange. The average bias measure is generally below 0.0002% across all

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turnover as daily share trading volume divided by tradable shares outstanding. The average daily turnover in China is 2.45%, indicating that it takes 1÷2.45%=41days to turn around all shares, which is much larger than the daily turnover of 1.11% in the U.S. during the same period. These summary statistics show that an average Chinese firm has smaller capitalization, higher valuation ratio, lower returns, and higher turnover than an average U.S. firm.

C. The Order Imbalance Measure

The order flows from different groups of investors are measured using order imbalances, as in Chordia and Subrahmanyam (2004). For stock *i,* day *d*, and investor group *G*, we compute

$$Oib_{i,d,G} = \frac{\sum_{j \in G} BuyVol_{i,d,j} - \sum_{j \in G} SellVol_{i,d,j}}{\sum_{j \in G} BuyVol_{i,d,j} + \sum_{j \in G} SellVol_{i,d,j}}, \tag{1}$$

where the numerator is the *difference* between buy and sell share volumes[7] for stock *i* on day *d*, summed over all individual *j*'s within group *G*, and the denominator is the *sum* of buy and sell share volumes for stock *i* on day *d*, summed over all individual *j*'s within group *G*. The data allows us to directly observe each trade's direction. When a set of investors buy more than they sell, the order imbalance is positive and vice versa. We compute the order imbalance measure for each investor group as *OibRT*1 to *OibRT*5, *OibINST,* and *OibCORP*.

Table I Panel C reports the summary statistics for the order imbalance measures. The average order imbalances for RT1 to RT5, institutions, and corporations range between -0.021 and 0.019, are all close to zero. The small magnitude of these average order imbalance measures indicates that most buys and sells within each investor group cancel each other out. The standard deviations of order imbalances are larger for large retail investors (0.352) and institutions (0.455),

stocks, which is negligible compared to the bias computed in Blume and Stambaugh (1983). Therefore, we compute daily returns using daily close prices without the Blume and Stambaugh (1983) adjustments.

[7] In this study, "share volume" refers to number of shares traded on that day, and "cash volume" refers to the product of share volume and the execution price.

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as compared to small and medium retail investors (between 0.166 and 0.250), indicate that there is more cross-stock variation in large retail investor and institutional trading activity. The one-day autocorrelation coefficient, AR1, for these *Oib* measures are 0.243, 0.259, 0.216, 0.059, and 0.102 for RT1 to RT5, respectively, suggesting that small and medium retail order imbalances are generally more persistent than large retail imbalances.

In terms of order flow correlations across the seven groups, the order flows from smaller retail investors, *OibRT*1, *OibRT*2, and *OibRT*3, are highly correlated, with correlation coefficients primarily higher than 0.60. *OibRT*4 is still positively correlated with *OibRT*1-*OibRT*3, but with lower correlation coefficients of around 0.20. The largest retail investors' order imbalance, *OibRT*5, is negatively correlated with all four other groups, with correlations around -0.15, indicating that this group of retail investors might have different trading patterns from the others. Institutional order imbalances are negatively correlated with all five retail groups, with correlations between -0.380 and -0.188, again implying different trading patterns from retail investors, even the largest retail investors. As we find earlier, corporations barely trade and their correlations with the rest of the investor categories are all lower than 10%.[8]

In this data section, we include *OibINST* and *OibCORP* to ensure the completeness of the summary statistics. Given that corporations are long-term investors and rarely trade, and that our study focuses on trading behavior, we exclude corporations from the remaining empirical results. In terms of institutional investors, given that retail investors are commonly assumed less sophisticated in comparison to institutional investors, we retain institutions in our main results for comparison purposes.

II. Can Retail Order Flows Predict Future Stock Returns?

[8] Appendix Figure I presents the time-series plot of the order imbalance measures for each investor group, and we find no evidence of time trends or breaks over our sample period.

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A. *Predicting Next Day Stock Returns Using Retail Order Flows*

To investigate the roles different retail investors play in the price discovery process, we first examine retail order flow's predictive power for the next day's return. We adopt the two-stage Fama and MacBeth (1973) regression to examine the predictive patterns of order flows for next-day returns. At the first stage, we estimate the following cross-sectional regression for each day *d* within each investor group *G*:

$$Ret_{i,d} = a0_{d,G} + a1_{d,G}Oib_{i,G,d-1} + a2_{d,G}{}'Controls_{i,d-1} + u1_{i,d}, \qquad (2)$$

where the dependent variable $Ret_{i,d}$ is the stock return for firm *i* on day *d*, and the independent variables include order imbalance measures from investor group *G* from the previous day, $Oib_{i,G,d-1}$, and control variables, $Controls_{i,d-1}$. We follow previous literature for the choice of control variables. To control for potential momentum/reversal from past returns, we include returns from the previous day, *Ret*(-1), returns from the previous week, *Ret*(-6,-2), and returns from the previous month, *Ret*(-27, -7). For size, value, and liquidity effects, we include log market size (*Size*), earnings-to-price ratio (*EP*), and turnover as controls, all computed from the previous month-end.

From the first stage estimation, we obtain the daily time series of coefficients for each investor group *G*, $\{a0_{d,G}, a1_{d,G}, a2'_{d,G}\}$. For the second stage estimation, we conduct statistical inference based on the means, $\{a0_G, a1_G, a2'_G\}$, and standard errors of the first stage coefficients. To be specific, we compute standard errors following the method in Newey and West (1987), with the optimal bandwidth chosen by using Newey and West (1994) approach.[9] If the order flow variable from a specific investor group *G* predicts future returns in the correct direction, defined

[9] We also consider robustness check using BIC to select optimal lag number. Results are similar and available on request.

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as an occasion when more past purchases are associated with higher future returns and more past sales are associated with lower future returns, we expect the coefficient $a1_G$ to be significantly positive, and vice versa.[10]

The estimation results for equation (2) are reported in Table II, which displays distinctive predictive patterns across different groups of retail investors. For the smallest retail investor group, RT1, the coefficient on the retail order flow variable is -0.0093, with a significant *t*-statistic of -17.36. The negative coefficient indicates that if retail investors RT1 buy more than they sell on a given day, the next-day return on that stock is significantly negative. To understand the economic magnitude of the coefficients, we report the interquartile range for *OibRT*1 at the bottom of the table. Multiplying the interquartile range, 0.2222, by the regression coefficient of -0.0093, generates an interquartile daily return difference of -21 basis points (more than 50% annualized!). The predictive patterns are qualitatively similar for retail investors in groups RT2-RT4. All coefficients are negative and statistically significant, and the daily interquartile return differences are -17, -11, and -2 basis points for RT2, RT3, and RT4, respectively. That is to say, the first four groups of investors trade in the incorrect direction versus future price movements: the stocks they buy have lower returns, and the stocks they sell have higher returns. It is possible that these retail investors are short of finance literacy, or their trading displays behavioral biases, which lead to the wrong return predictions. Interestingly, when we move from smaller account sizes to larger ones, the negative coefficients become smaller, indicating that larger retail investors trade more correctly than smaller ones. Indeed, for the largest retail investors, RT5, the coefficient on the past day order imbalance is 0.0012, which is positive and significant with a *t*-statistic of 10.48. The interquartile

[10] As an alternative to the Fama-Macbeth regression, we also adopt the portfolio sorting approach, using order imbalance measures from different groups as sorting variables. Results are similar to the results of the Fama-Macbeth regression, and are discussed in robustness check in Section IV.D.

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daily return difference is five basis points per day (over 12% per year). It appears that the trading of the largest retail investors predicts the cross-section of future stock price movements in the correct direction.

[Insert Table II about here]

For comparison, the coefficient on the previous day order imbalance is 0.0016 for institutions, with a *t*-statistic of 18.06. That is to say, institutional order flows predict future stock price movements in the correct direction, and the interquartile return difference is 10 basis points per day (over 25% per year), which is approximately twice the magnitude of the RT5 estimate. This finding is consistent with many previous studies showing that institutional investors are generally more informed than retail investors.[11]

For the control variables, the coefficients on the previous day return have mixed signs, while the coefficients on the previous week and previous month returns are all negative and significant, indicating strong reversals over the weekly and monthly horizons. Size is mostly insignificant, whereas the earnings-to-price variables are always positive and significant, indicating a strong value effect. The coefficients on turnover are always negative and significant, suggesting that higher turnover leads to lower returns in the future. These findings are consistent with those of previous studies on the Chinese stock market, such as Liu et al. (2019). These results also confirm that the predictive power of the various order flow variables for future stock returns is not a manifestation of the size, value, liquidity, or momentum/reversal effect.

B. Predictive Patterns with Different Counter-parties

[11] Readers might wonder whether the RT5 are corporate insiders, which allow them to have inside information and positive prediction for future returns. China has strict rules on the windows during which insiders can trade related stocks. For instance, they cannot trade before significant corporate announcements, such as earnings announcements. Later results show that RT5 trade actively around earnings announcements, which support the notion that they are unlikely to be corporate insiders.

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Trading dynamics among counterparties is an interesting and important concept that helps to understand how different groups of investors interact with each other. However, most existing studies cannot offer much insight into this because it is difficult to pin down each trade's counterparties, due to data limitation.[12] We are fortunate to have the whole trading history from the exchange, which makes it possible for us to identify the counterparties' group for each trade. Therefore, in this subsection, we closely examine the trading dynamics among different groups of investors, and whether different trading pairs are associated with different return predictive patterns.

We group trades by counterparties from the buy and sell sides of each trade in three steps. First, because corporations rarely trade and account for less than 2% of daily trading volume, we exclude trades with corporations. Second, we regroup the remainder of the six groups into three larger groups to reduce overall dimensions: 1) RT1 to RT4 are bundled together as one group because their order imbalances are positively correlated, and they share similar predictive patterns; and 2) RT5 and INST are considered as two separate groups because their order flows are negatively correlated with those from smaller retail investors, and the order flows from RT5 and INST are also negatively correlated. For the final step, with three groups of investors and two sides of trades (buy or sell), we divide all the remaining trades into six bins: BBS, BSB, BSS, SBB, SBS, and SSB. The first letter indicates the trade direction of RT1-RT4, the second indicates the trade direction of RT5, and the third indicates INST. For instance, "BBS" means RT1-RT4 "buy", RT5 "buy" and INST "sell". Table III Panel A presents the proportion of each type of trade. The highest

[12] Exceptions include Boehmer, Sang, and Zhang (2021), which examines the trading patterns of retail investors following insider trading using retail trading trades identified from TAQ and Thomson Reuters Insiders data; and Van Kervel and Menkveld (2019), which examines the high-frequency trading around large institutional orders. Unlike the rest of the studies, these two papers can identify the counterparties of the trades in their sample, but their samples only cover limited subsets of the market.

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proportions are "BSS" and "SBB", both above 20%, where RT5 is on the same side as institutions. The lowest proportions are for "BSB" and "SBS", both lower than 13%, indicating that RT1-RT4 are less likely to be on the same side as institutions.

To address whether the return predictive patterns change when the counterparties are different, we modify equation (2) to create equation (3), as follows:

$$Ret_{i,d} = b0_{d,G} + \left(\sum_{k=1}^{6} b1_{k,d,G}\, I_{k,d}\right) Oib_{i,d-1,G} + b2'_{d,G} Controls_{i,d-1} + u2_{i,d}. \quad (3)$$

According to the six bins, we define the corresponding indicator variable $I_{k,d}$, which takes the value of one when the trade belongs to the *k*-th bin, and zero otherwise. We estimate equation (3) for each of the three investor groups, RT1-RT4, RT5, and INST, using the Fama-MacBeth regression. The first stage coefficient $b1_{k,d,G}$ captures the predictive power for the *k*-th combination on day *d* for investor group *G*, and the second stage coefficient $b1_{k,G}$ (the mean of the first stage time-series of coefficient) measures the average predictive power for the *k*-th combination for group *G*.

[Insert Table III about here]

Panel B of Table III reports the estimation results. Our prior is that given that the smaller investors, RT1-RT4, predict returns negatively, they might be at a disadvantage while trading against the large retail investors and the institutional investors. We focus on the order flows from RT1-RT4 in the first specification. The coefficients on the order flows from RT1-RT4 are consistently negative, implying that small retail investors' negative predictive power is prevalent and does not depend on counterparties. We notice that the coefficient magnitudes are relatively large when RT1-RT4 are on the opposite side of both RT5 (BSS) and INST (SBB), which supports our prior that RT1-RT4 might be at a disadvantage when trading against both RT5 and INST. In specification II, we focus on the predictive pattern of RT5 with different counterparties. The

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coefficients are mostly positive, consistent with RT5's positive predictive power for returns in general. Interestingly, the coefficients turn to negative for the cases of BBS and SSB, when RT5 are on the same side of RT1-RT4, but on the opposite side of INST. This suggests that when RT5 sides with smaller retail investors, and trades against INST, their prediction becomes erroneous. Similar patterns also apply to INST in regression III, which has positive coefficients when INST trades on the same side of RT5, but the coefficients become negative when INST trades on the opposite side of RT5. That is, when RT5 and INST agree with each other, and trade against the smaller retail investors, their prediction for future return is correct, but when RT5/INST agree with the smaller retail investors, but disagree with each other, they have the incorrect signs for predicting future returns.[13]

C. Predictive Patterns with Different Holding Horizons

Another key variable for investment is the holding horizon. The statistics in Table I show that Chinese retail investors have high turnover ratios, which lead to short holding horizons. Does the predictive power of order flows change when holding horizons vary? Conventional wisdom suggests that shorter holding horizons are likely to be associated with excessive trading, and possibly poorer returns. Due to data limitation, it is difficult for most existing studies to directly verify how holding horizons are associated with predictive power for a large sample. Taking advantage of our data, we compute holding horizons for each investor group, and examine how holding horizons are related to predictive patterns of retail order flow.

The first step is to find a measure to capture holding horizons. Direct and accurate holding horizon calculation is quite challenging, as there are partial orders, cross trading, and we need to make assumptions regarding the in and out orders. To overcome this, we borrow the simple and

[13] We also form portfolios based on order imbalance and trading counterparties, and present the results in Appendix Table IV Panel A. The results are qualitatively similar to these in Table III.

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intuitive days-to-cover-ratio (DTCR) idea from the short selling literature, which measures the number of days it takes for all short shares to be covered, as in Diether, Lee, and Werner (2008) and Boehmer and Wu (2013). That is, we define the holding period of average type G investors for stock i on day d as $DTCR_{i,d,G} = \frac{TotalHoldingShares_{i,d,G}}{DailyShareVolume_{i,d,G}}$, or the total shares held by investor group G divided by the daily shares traded on stock i for type G investors. For example, if 20% of the total shares are held by RT5, and the RT5 daily trading volume for this stock is 1%, then it takes 20 days for the entire holding stock by group G to be covered, and the days-to-cover-ratio for RT5 would be 20 days. Panel A of Table IV reports the cross-sectional distribution of DTCRs for each investor group. Overall, the DTCRs are similar among RT1-RT4, with means of approximately 50 days, whereas RT5 and INST have longer horizons of approximately 200 days.

To examine how DTCR, a rough measure for holding horizons, is associated with the return predictive pattern, we simplify our analysis by selecting three benchmark horizons: 10, 20, and 60 days, which roughly correspond to the 25th, 50th and 75th DTCRs, across different groups. Next, we separate all stock-day-investor group observations, depending on the DTCR calculated from the previous day, into four bins: (0,10], (10,20], (20,60], and above 60 days. We modify equation (2) to create equation (4), which incorporates the differences in DTCR as follows:

$$Ret_{i,d} = c0_{d,G} + \left(\sum\nolimits_{k=1}^{4} c1_{k,d,G} I_{k,d}\right) Oib_{i,d-1,G} + c2'_{d,G} Controls_{i,d-1} + u3_{i,d}. \qquad (4)$$

Here, the indicator $I_{k,d}$ takes the value of one if the trade belongs to the k-th bin of DTCR, and zero otherwise. If conventional wisdom is correct, shorter holding horizons may be associated with poorer predictive power. Since the DTCR measure is typically correlated with trading volumes, which can have predictive power for returns by itself, as in Gervais, Kaniel, and Mingelgrin (2001), we include a volume variable as an additional control variable. In particular, we follow Gervais et

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al. (2001) and construct the relative trading volume (*Rvol*) measure, which is the trading volume on the stock-day divided by the average daily stock trading volume from previous 49 days.[14] Other control variables are the same as those in equation (2).

[Insert Table IV about here]

The estimates of $c1_{k,G}$ are presented in Panel B of Table IV. Take RT1 as an example. For the short horizon of less than 10 days, the coefficient is -0.0171 and highly significant. When the holding horizon become longer, the coefficient slightly decreases in magnitude to -0.0086 for 10 to 20 days, -0.0064 for 20 to 60 days, and -0.0074 for longer than 60 days. This decay shows that mistakes in trading direction of the smallest retail investors slowly diminish when holding horizons become longer. The changes are more dramatic for RT2 to RT4, which begin with negative and significant coefficients for shorter horizons, and eventually become mostly insignificant for horizons longer than 60 days. These findings are consistent with the conventional wisdom that shorter holding horizons are likely associated with poorer returns, while longer horizons improve performances. Interestingly, for RT5, the patterns are different. For horizons shorter than 10 days, the $c1$ coefficient is 0.0056 and highly significant, but the $c1$ coefficient quickly reduces to 0.0015 (still highly significant) for holding horizons of 10 to 20 days, and eventually becomes insignificant for holding horizons longer than 60 days. One possibility for the decreasing coefficients and statistical significance is that RT5 might trade on time-sensitive information, which is gradually incorporated into price, leading to the pattern of strong predictive power of shorter holding horizons than longer holding horizons.

The pattern for INST is somewhat similar to that for RT5, whereas the rate of reduction for the coefficients is much slower. In fact, for horizons shorter than 10 days, the $c1$ coefficient is

[14] We thank our referee for this suggestion.

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0.0017, and it slowly decreases to 0.0008 for holding horizons longer than 60 days. In this case, it is possible that the INST's order flow contains some time-sensitive information, which explains the reduction; it is also possible that the INST order flow contains information related to firm fundamentals, which takes longer for the information to be revealed and thus, *slow* decay. Of course, most of these coefficients are positive and significant, indicating that, even for longer holding horizons, the order flows from RT5 and INST still contain price-relevant information in the right direction.[15]

D. Predicting Long Term Stock Returns Using Retail Order Flows

Previous exercises focus on next-day return prediction, and it is natural to ask whether the predictive patterns continue for longer terms. If the predictive pattern quickly vanishes or reverses, the return predictability may be driven by short-term noise. If the predictive pattern persists over longer horizons, the return predictability is more likely to be linked to firm fundamentals or persistent biases. Therefore, we extend the Fama-MacBeth specification in equation (2) to create equation (5), where longer horizons are up to 12 weeks:

$$Ret_{i,w} = d0_{w,G} + d1_{w,G} Oib_{i,d-1,G} + d2'_{w,G} Controls_{i,d-1} + u4_{i,w}, \tag{5}$$

That is, we use the previous day order imbalance from group G, $Oib_{i,d-1,G}$, to predict weekly returns over the next w-th week. To be more specific, $Ret_{i,w}$ is calculated as a 5-day return from the beginning to the end of week w, where w=1, …, 12. For instance, when w=1, $Ret_{i,w}$ is the cumulative return over days d to d+4, and when w=12, $Ret_{i,w}$ is the cumulative return over day d+55 to d+59. If order imbalances have only short-lived predictive power for future returns, we should observe the coefficient $d1$ quickly reverses. Alternatively, if the specified retail order

[15] As an alternative to the Fama-MacBeth regression, we construct double sort portfolios based on order imbalance and holding horizons. The results are reported in Appendix Table IV Panel B, and are qualitatively similar to those in Table IV.

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imbalance has a longer predictive power, the coefficient $d1$ may slowly decrease, rather than quickly reverse.

[Insert Table V about here]

We present the estimates of coefficient $d1$ in equation (5) in Table V. For the smallest retail investor RT1, the coefficients on $OibRT1$ are negative and monotonically increase from -0.0226 at week one to -0.0005 at week 12, while the coefficients are statistically significant until week 9. Reverse patterns are not observed, which implies that the predictive power is long lasting. Similar patterns are observed for $OibRT2$ and $OibRT3$. For $OibRT4$, the coefficient is -0.0019 and highly significant at week 1, but quickly reverses and becomes insignificant, indicating that its predictive power for future returns might be temporary. The positive predictive power of $OibRT5$ and *OibINST* also significantly persist for approximately 8 to 9 weeks, and there are no significant reversals. The general persistence of the positive predictive pattern for RT5 and INST indicates that the predictive power is likely rooted in information related to fundamentals, or against persistent noise/behavioral biases.[16]

III. What Drives the Order Imbalance Predictive Power for Future Returns?

A. *Alternative Hypotheses for Explaining Retail Order Flow and Its Return Predictive Power*

Given the large differences in the predictive power of future returns for different investor groups' order flows, it is important to understand the driving forces behind these order flows. Previous literature provides several hypotheses for explaining investor order flows, which may help to explain the heterogeneous predictive patterns of different investor groups for future returns. Below we outline four main hypotheses, and these hypotheses are not mutually exclusive.

[16] We also consider an alternative specification for the long-term predictability. To match the weekly returns on the left-hand side, we use the previous week order imbalance $Oib_{i,w-1,G}$, to predict future weekly returns. The results are reported in Appendix Table V are similar to those in Table V.

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First, Chordia and Subrahmanyam (2004) state that order flows tend to be persistent, and that persistent buying/selling pressure could directly lead to the predictability of future returns. Here we adopt the previous day order imbalance measure, $Oib_{i,d-1,G}$, as the proxy for order persistence.

Second, Kaniel et al. (2008) argue that retail traders in the U.S. are mostly contrarian, providing liquidity to the market, and afterwards receiving positive premium for liquidity provision. Following this logic, if the retail trades are momentum trades that demand liquidity, it is then possible that momentum trades will have a negative relation with future returns; and if the retail trades are contrarian and provide liquidity, these contrarian trades will have a positive relation with future returns. For this hypothesis, we choose returns from the previous day, week, and month as proxies for momentum or contrarian trading.

Third, Liu et al. (2021) connect retail trading motives to behavioral biases and find that overconfidence and gambling preferences are the two dominant behavioral biases that affect trade of Chinese retail investors. For the overconfidence measure, we follow Barber et al. (2009) and Liu et al. (2021), and proxy it with the average of the daily investor group's stock turnover from previous 20 days, $Overconf_{i,d-1,G}$.[17] For gambling preferences at the stock level, $Gamble_{i,d-1}$, we follow Bali et al. (2011) and compute the maximum daily returns from the previous 20 days.[18]

[17] For the overconfidence proxy, Barber and Odean (2000) use each household's portfolio's turnover. Our data has a different structure and cannot be used to construct household level proxies. Considering our study focuses on investor groups rather than individual households, we carry the spirit from the previous literature and use the stock-level turnover from each investor group.

[18] An alternative measure for gambling preference proxy is introduced in Liu et al. (2021), which relies on events when the stock return hits a 10% price limit. However, the 10% price limit only accounts for 0.07% of the total sample, and is not suitable for our purposes using on daily*stock returns. Thus, we choose the maximum daily return as our main gambling measure. We also consider other alternative proxies for gambling preferences, such as idiosyncratic volatility and skewness. These proxies deliver similar results to those using maximum daily returns, and are available upon request.

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Finally, Kelley and Tetlock (2013) find that retail investors, especially the aggressive ones, may have valuable information about fundamental firm news, and thus, their trading could correctly predict the direction of future returns. We measure firm-level fundamental news by the cumulative abnormal returns (*CAR*) over the earnings announcement period. Unlike the proxies for order persistence, liquidity provision, and behavioral biases, which can be computed for each stock on each day, the news proxies are only available on earnings news days, which account for 1.58% of stock days, rendering our two-stage estimation (to be introduced in Section III.B.) imprecise. To cope with this missing data issue for the news hypothesis, we first consider the order persistence, liquidity provision, and behavioral bias hypotheses in Section III.B, and focus on the news hypothesis using an event-day approach in Section III.C.

B. A Two-Stage Decomposition for Order Flows' Return Predictive Power

To find out whether the above hypotheses explain the trading behavior of different retail investor groups and their predictive power for future stock returns, we adopt a two-stage decomposition method as in Boehmer et al. (2021).[19] For the first stage, we use the above hypotheses to explain the retail flow measures to find out which ones are important drivers for the order flows. This step also helps to decompose the retail order flows into hypothesis-implied components for each hypothesis. For the second stage, we investigate which of the hypothesis-implied components contributes to the predictive pattern of different investor order flow measures.

Table VI Panel A reports the first-stage estimation results. In the first row, the coefficients of the lagged order flow variables are always positive and significant, indicating that order persistence is an important driver of order flow. For the next three rows, order flows are connected with returns from the previous day, week, and month. The order imbalances of RT1, RT2, and RT3

[19] A step-by-step description of the two-stage decomposition is provided in Appendix A.

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load positively and significantly on the previous day's return, indicating that these investors buy more if the previous day's return is positive, and sell more if the previous day's return is negative. This corresponds to a daily momentum trading strategy, which demands immediate liquidity. For larger retail investors in RT4 and RT5, order imbalances load negatively and significantly on returns from the previous day, indicating that they are contrarian investors who buy low, sell high, and possibly provide immediate liquidity. If we extend the horizon to the previous one week or one month, the coefficients on all returns are negative and significant, indicating that all retail investors become contrarian and buy losers, and sell winners over the longer term.[20]

[Insert Table VI about here]

The next two rows present the results on how behavioral biases are related to order flows. The coefficients on the overconfidence proxy are all positive and significant for RT1-RT4, indicating that overconfidence might be a strong driver of trading by these retail investors. For the largest retail group, RT5, the coefficient becomes -0.0881, with a significant *t*-stat of -8.80. In other words, the largest retail investors' trades are in the opposite direction of the overconfidence proxy. In terms of gambling preference, for RT2-RT4, the coefficients are always positive and significant, indicating that these retail investors prefer to buy stocks with lottery features.[21] When

[20] Our finding that large retail investors are contrarian and smaller ones are momentum traders over daily horizon differs from some previous studies. For instance, contrarian patterns are documented in Kaniel et al. (2008) using monthly horizons in the U.S., and Barrot et al. (2016) using daily and weekly horizons in France. Using U.S. data, Kelley and Tetlock (2013) and Boehmer et al. (2021) both find that retail trades follow momentum over daily horizons, but are contrarian at weekly horizons. In our setting, we find the trading patterns from investors with smaller account sizes are similar to those in Kelley and Tetlock (2013) and Boehmer et al. (2021), while the investors with the largest account sizes behave similarly to the patterns documented in Kaniel et al. (2008) and Barrot et al. (2016). We also examine whether the momentum/contrarian patterns are potentially related to average holding periods of stock. Interacting the return of the previous day with holding horizons, we find small retail investors' momentum are stronger for short holding periods, while large retail investors' contrarian is similar for both short and long holding periods. The results are available on request.

[21] The increasing coefficients from RT1 to RT4 should not be interpreted as RT4's gambling preference being stronger than RT1, because these variables are not standardized and cannot be compared directly. If we standardize them and make them comparable across investor groups, RT3 has the strongest gambling preference. The results are available on request.

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we move on to RT5, the coefficient is -0.0671 with a significant *t*-stat of -2.53, which indicates that the largest retail investors' trades are in the opposite direction of the gambling motive.

We report the second stage decomposition results in Panel B of Table VI. We consider the first retail group, RT1, as an example. The coefficient estimate on *Oib(Persistence)* is -0.0301, with a *t*-statistic of -16.73, which implies that order persistence significantly and negatively contributes to the predictive power of the RT1 trading flow. The coefficient estimate on *Oib(Liquidity)* is -0.0135, with a *t*-statistic of -4.48, which suggests that daily momentum trading probably significantly and negatively contributes to the predictive power of the RT1 trading flow. The coefficient of *Oib(Overconf)* is -0.1006, with a *t*-statistic of -4.00, and the coefficient of *Oib(Gamble)* is -0.2877, with a *t*-statistic of -4.89. These two significant coefficients imply that the two behavioral biases significantly and negatively contribute to the predictive power of RT1 trading flow. For the *Oib(Other)* component, the coefficient is -0.0086, with a significant *t*-statistic of -15.54, indicating that there is residual information other than those incorporated in the three hypotheses, which significantly contributes to RT1's negative predictive pattern for future returns. In terms of the economic magnitude, we compute the interquartile range of all five components of the order imbalance measure. For the smallest retail group RT1, if we move from the 25^{th} percentile to the 75^{th} percentile in the distribution, the interquartile differences in future one-day stock returns are -0.1177%, -0.0290%, -0.0337%, -0.0314%, and-0.1778% for *Oib(Persistence)*, *Oib(Liquidity)*, *Oib(*Overconf*)*, *Oib(Gamble)*, and *Oib(Other)*, respectively. In other words, order persistence, liquidity demand, overconfidence, and gambling preferences all contribute to the negative predictive power of RT1 for next-day returns, with the first term having the largest magnitude. Similar patterns are observed for other smaller retail investor groups RT2-RT4.

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Regarding the largest retail investors, RT5, the patterns are quite different. In terms of the coefficient estimates, we find that order persistence, overconfidence, gambling preferences, and others are all positive and significant. In terms of economic magnitude, if we move from the 25^{th} percentile to the 75^{th} percentile in the distribution, the interquartile differences in future one-day stock returns are 0.0281%, 0.0097%, 0.0257%, 0.0425%, and 0.0490% for *Oib(Persistence)*, *Oib(Liquidity)*, *Oib(*Overconf*)*, *Oib(Gamble)*, and *Oib(Other)*, respectively. This indicates that order persistence, liquidity provision, and trading against behavioral biases all contribute to RT5's positive predictive pattern for future returns.

Overall, our decomposition exercise shows that a substantial part of the negative predictive power of retail investors with smaller account sizes comes from order persistence, liquidity demand, and behavioral biases, while the positive predictive power of retail investors with larger account balances mostly comes from order persistence and trading against overconfidence and gambling preferences.[22] Across all investor groups, the significance and large magnitude of the "other" component indicate that existing hypotheses cannot fully explain trading behaviors and their predictive power for returns. Then, what does "other" stand for? One possibility is information, which we take a close look at in the next subsection.

C. A Close Look at the Information Channel

It is important to understand how various retail investors participate in the information discovery process. As mentioned earlier, the most influential information at the firm level is earnings news, hence we follow Kelley and Tetlock (2013) and measure firm-level information by the cumulative abnormal returns (*CAR*) over the earnings announcement period. Notice that

[22] In Appendix Table VI, we examine the robustness of two-stage decomposition by adding additional two weeks lag of order flow in the first stage, construct the new "persistence" component, and re-estimate the second stage decomposition. The results are robust.

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earnings news only happens quarterly rather than daily, so the daily Fama and MacBeth (1973) estimation adopted for the two-stage estimation might not be proper for understanding how Chinese retail investors process information. As an alternative, in this section, we focus on event-days to study this issue. We capture each retail investor group's participation in the information discovery process in three steps.

First, we examine whether different retail investors can predict earnings news the next day. A positive answer indicates that these investors anticipate the information before it becomes public, either because they have access to private information or because they have better skills in prediction. We estimate the following cross-sectional specification for each quarter q:

$$CAR_{i,d-1,d} = e0_q + e1_q Oib_{i,d-1} + e2'_q Controls_{i,q-1} + u5_{i,q}. \quad (6)$$

Assuming that the earnings announcement day is day d, we compute the cumulative returns over day d-1 and day d and subtract the market returns over the same period to obtain cumulative abnormal returns for each stock, $CAR_{i,d-1,d}$.[23] The main predictive variable on the right-hand side of the equation is the order imbalance measure from day d-2. Notice that each firm has only one earnings day per quarter, and equation (6) is estimated for each quarter in the cross-section to ensure that we cover all firms in each quarter. Statistical inferences are based on the quarterly time-series of the estimated coefficients, and standard errors are computed using Newey-West (1987) method with the optimal bandwidth chosen by using Newey and West (1994) approach. If retail order flows can predict earnings surprises in the right direction, coefficient $e1$ should be significantly positive, and vice versa. Panel A of Table VII presents the estimation results. For retail investors RT1-RT3, the coefficients $e1$ are -0.0251, -0.0234, and -0.0166, respectively, with highly significant t-statistics. These negative and significant coefficients indicate that these

[23] We also examine wider window such as CAR(-1,1) and CAR(-3,3), the results are similar and available on request.

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investors incorrectly predict earnings surprises. Coefficient $e1$ for RT4 is close to zero and insignificant. In contrast, the coefficient $e1$ for RT5 is 0.0023, positive and statistically significant, implying that these investors can correctly predict future earnings surprises.

[Insert Table VII about here]

Second, we examine whether different retail groups can process contemporaneous public news. Here, the dependent variable is retail order flow, $Oib_{i,d}$, which we connect to contemporaneous earnings news, $CAR_{i,d-1,d}$:

$$Oib_{i,d} = f0_q + f1_q CAR_{i,d-1,d} + f2'_q Controls_{i,d-1} + u6_{i,q}. \tag{7}$$

If a particular type of retail order imbalance can process contemporaneous and public earnings news in the right direction, the associated coefficient $f1$ is significantly positive and vice versa. Panel B of Table VII reports these results. For retail investors RT1-RT4, the coefficients $f1$ are -1.9225, -1.8291, -1.4349, and -0.8781, respectively, with highly significant *t*-statistics. These negative and significant coefficients indicate that these retail investor groups process contemporaneous public earnings news in the wrong direction. By contrast, the coefficient $f1$ for RT5 is 0.1583, with a *t*-statistic of 2.12, implying that RT5 might be able to correctly process contemporaneous public earnings news.

Third, we examine whether the predictive power of retail order flows for future returns improves or deteriorates on event days to understand how much the information hypothesis helps explain the return predictive patterns observed in Section II. Here, we add the event-day dummy and interaction term:

$$\begin{aligned} Ret_{i,d} = g0_d + \left(g1_d + g2_d Event_{i,d-1}\right) Oib_{i,d-1} + g3_d Event_{i,d-1} \\ + g4'_d Controls_{i,d-1} + u7_{i,d}. \end{aligned} \tag{8}$$

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The event dummy $Event_{i,d-1}$, is equal to one if firm i has earnings news on day d-1 and zero otherwise. For non-news days, the predictive power of retail trades is measured by the coefficient $g1$; for news days, the predictive power is measured by ($g1$+$g2$). If coefficient $g2$ is significantly different from zero, the group of retail investors anticipates future stock returns differently on these news days. In the U.S., firm earnings announcements are chosen by firms and are scattered throughout the year. In China, all firms are required to report their financial statements to regulators before the four preset deadline dates each year. Consequently, firms mostly announce their earnings within a short period before these deadline dates, and there would be zero announcements outside these short periods. To ensure that we have enough observations to estimate the Fama and Macbeth (1973) coefficients in equation (8), we only include days with at least 5% of the total number of firms with earnings announcements, which gives us 68 days, or 8% of the total days in our sample.

Table VII and Panel C present the results. We consider the smallest retail investor, RT1, as an example. The coefficient on order imbalance, $g1$, is -0.0079 and statistically significant, indicating that, on average, the trades from RT1 negatively predict future returns. When there is earnings announcement news, the coefficient on the interaction of the event dummy and the order imbalance is -0.0080 and is statistically significant, implying that the negative prediction of RT1 for future stock returns doubles on earnings news days. This is consistent with our earlier finding that smaller retail investors fail to predict and process earnings news, which leads to more negative predictions of returns on event days. Similar patterns are observed in RT2, RT3, and RT4. For the largest retail investor, RT5, the coefficients $g1$ and $g2$ are 0.0005 and 0.0014, respectively, both of which are statistically significant. In other words, large retail investors' predictive power for future

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returns quadruples on earnings news days, possibly because these retail investors can correctly predict and process earnings news, which enhances their ability to predict future stock returns.

Overall, our results reveal interesting heterogeneous patterns in how retail investors predict and process public information. Smaller retail investors are unable to predict future news and lack the skills to correctly process public news, while the largest retail investors and institutions can correctly anticipate future earnings news and incorporate contemporaneous news into their trading. The differences in the information-processing abilities of different retail investors clearly contribute to the differences in their predictive power for future returns.[24]

IV. *Trading Performances Using Tracking Portfolios of Retail Trading*

A. The BLLO Method

With the short- and long-term predictive patterns documented in the previous sections, it is natural to ask how good the trading performance would be if we track the order flows from different groups of retail investors. To answer this question, we follow Barber, Lee, Liu, and Odean (2009, BLLO hereafter), which designs a simple and intuitive method to compute the trading performances of different types of investors in Taiwan Stock Exchange over 1995 to 1999. A step-by-step description of BLLO's method is provided in Appendix B. Here we provide a brief description of the BLLO's method, which has four steps.

First, BLLO separate all investors into "individuals", "corporations", "dealers", "foreigners", and "mutual funds", using identities provided by the exchange. Second, within each investor group, BLLO compute the aggregate daily net buy and net sell positions and create daily matching trading portfolios, the "net buy" and "net sell" portfolios. Third, BLLO track these "net

[24] As an alternative, we use the Financial News Database of Chinese Listed Companies (CFND) to investigate whether the results from earnings news can be extended to other news. The results confirm our findings in Section III, and are available on request.

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buy" and "net sell" portfolios over a holding horizon of *n* days, and compute cumulative cash flows and returns from this tracking strategy, and treat them as proxies for trading performances of different investor groups. After computing the total performance, BLLO decomposes the total into three intuitive components, stock selection, market timing, and trading cost.

We adopt the BLLO method in our study for three reasons. First, the data structure of our sample is quite similar to those of BLLO's, therefore it is relatively straightforward to apply to our data. Second, the BLLO method is simple and easy to interpret, and it allows us to compare our results with their results. Third, the decomposition of total performance into stock selection, market timing, and trading cost helps us to connect our earlier results on the predictive power of order flows in the cross section. Notice that the performance estimates from the BLLO method are not realized gains and losses from investors. Rather, they are estimates of a trading strategy that follows signals from retail trading, namely, the net positions of each investor group, which reflects the investment skills of different groups of investors.

We make three parameter choices within the BLLO framework. First, BLLO assumes that the net buy and sell portfolios are held for *n* days and sets *n* to different horizons. As the main discussion of BLLO focuses on the holding horizon of 140 days (n=140), we choose the same horizon for ease of comparison. Second, BLLO assumes that the net buy (sell) portfolios are independent for each day *d*, and annualizes the performance by aggregating the daily net buy and sell portfolios over each year. Here we relax the independent assumption, and compute the standard error estimates using the Newey and West (1987) method, with the optimal bandwidth chosen by using Newey and West (1994) procedure. [25] Third, to have a perspective of total performance as a percentage of the total investment of these tracking portfolios, which is similar to the idea of return

[25] We conduct robustness check using optimal lag numbers from Bayesian Information Criterion (BIC). The results are similar and available on request.

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on investments, BLLO measures total investment as the aggregate holding value for each group of investors. For the ease of comparison, we choose the aggregate holding values in Panel A of Table I and present the return on investment as the total performance over aggregate holding.

B. Performances of Chinese Investors' Trading Using the BLLO Method

Panel A of Table VIII reports the performance of the tracking portfolios. The annualized return to investment ranges between -5.61% and -2.80% for a 140-day holding horizon for RT1 to RT4, and RT5 have a close to zero, -0.29%, return to investment, while the institutions have a return to investment of 1.15%. It might not be too surprising to find that BLLO's net buy and sell tracking portfolio approach provides negative returns on investments for RT1-RT4, given that our earlier results show that RT1-RT4 predict future returns negatively and significantly. Similarly, positive returns on investments for INST are also expected because earlier evidence shows that institutional order flows predict future returns positively and significantly. It is intriguing to find that the net buy and sell portfolios tracking the order flows of RT5 deliver negative returns on investments, while earlier results show that RT5 predicts future returns positively and significantly.

The answer is in the next three columns, which provide the three components of total performance. For RT5's annualized return to investment of -0.29%, the stock selection component contributes a significant and positive coefficient of 1.05%, which is consistent with the earlier positive cross-sectional return predictive power; the market timing component contributes an insignificant coefficient of -0.30%, indicating that RT5 might not be able to time the market; and the trading cost component contributes a significant coefficient of -1.03%. That is, even though RT5 is capable of stock selection, their low market timing ability and significant transaction cost cause the tracking portfolio to have a close-to-zero total return on investment.

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The decomposition of RT1-RT4 is different from that of RT5 in the sense that their groups of investors do not possess either stock selection or market timing ability, and so all three components of total returns on investment are all negative, mostly attributable to poor stock selection and significant trading cost. For instance, the trading cost component ranges between -1.38% (RT1) and -1.80% (RT2), which suggests that RT2 pays the most for trading relative to their holding. In the case of institutional investors, the results are quite similar to RT5, except that they have superior stock selection ability (StkSelect=1.38%), market timing ability (MktTiming=0.19%), and lower trading costs (TrdCost=-0.41%), and the total return on investment is positive and significant. [26]

[Insert Table VIII about here]

C. Performances for Momentum and Contrarian Trading Using the BLLO Method

The previous section shows that different retail investors have different momentum and contrarian patterns, which contribute to their different predictive patterns for future returns. For instance, smaller retail investors display daily momentum patterns, which contribute to their negative return predictive pattern, while the largest retail investors display daily contrarian trading patterns, which contribute to their positive predictive pattern for future returns. Does the momentum or contrarian trading pattern also play a significant role in the performance of net buy- and sell-tracking portfolios?

To answer this question, we separate the total performances in equation (12) into two scenarios, "daily momentum" and "daily contrarian". If the investor group G's order imbalance times previous day return, $Oib(i,d,G) * Ret(i,d-1)$, is positive, we classify the stock i on day

[26] We compare Table VIII Panel A with BLLO's results for the Taiwan stock market and find that the general patterns are similar. In Appendix Figure II, we also conduct a similar exercise for shorter holding periods. The results display similar patterns as in Panel A of Table VI.

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d for investor group G to be "daily momentum" because they buy when price rises and they sell when price drops; if $Oib(i,d,G) * Ret(i,d-1)$ is negative, then we classify the stock i on day d for investor group G to be "daily contrarian". We then calculate the performance for each scenario.

Panel B of Table VIII reports the results. Taking RT1 for example, following the daily momentum pattern, the annualized return on investment for RT1 is -4.57% with *t*-statistic of -8.47, which is significantly worse than the annualized return on investment of RT1 at -1.04% with *t*-statistic of -2.57 when they follow contrarian strategy. The results show that momentum trading results in worse performance than that of contrarian trading. Similar patterns exist for RT2-RT4. For RT5, the annualized return on investment is -1.61% when their trades display daily momentum patterns and 1.32% when they display daily contrarian patterns. For INST, the annualized performances are positive in both momentum (0.51%) and contrarian (0.64%) conditions, while the contrarian conditions have slightly higher returns on investments. Regarding the three components of the total performances, the stock selection performances are in general better under the "daily contrarian" column than "daily momentum" column; the market timing component are mostly insignificant under "daily momentum", but significantly negative under "daily contrarian", the trading costs are similar for momentum and contrarian columns. In other words, investors generally perform better when they pursue daily contrarian rather than daily momentum trading strategies.

D. Performances around Earnings Announcements Using the BLLO Method

Given that earnings announcements are the most informative firm-level events, we are also curious to understand how they contribute to the overall performance of the tracking portfolios. Our prior finding is that if investors have an information advantage around earnings announcements, they would have better performance around earnings announcements, while if

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investors have disadvantages around these event days, they would have worse losses on these days. There are 15,702 quarterly earnings announcement events during our 3.5-year sample. As in Section III.C, we consider both the day before earnings announcements and the announcement day as event days, accounting for 2.98% (=15,702*2/1,053,148) of the total stock-day observations. For this analysis, we compute the total performance on these event days and compare it with the overall performance on all days.

We present the results in Table VIII Panel C. For ease of comparison, we present the estimates over the earnings event days under "EA", and its proportion to all days under "EA/all days". The annualized total performance over earnings days for RT1 is -0.39%, which accounts for 6.89% of the total performance of all days. The stock selection, market timing, and trading cost components are -0.36%, 0.02%, and -0.04%, accounting for 8.96%, -8.63%, and 3.06% of all days, respectively. The coefficients on total performance, stock selection, and trading costs are all statistically significant. Considering that earnings days only account for 2.98% of all days, the total performance and stock selection component of the tracking portfolio following RT1 order flows is particularly bad on earnings days, which echoes our findings in Section III. The trading cost on EA days accounts for approximately 3% of all days, which is close to the overall number of EA days. Similar patterns are observed for RT2-RT4. For RT5, the annualized EA total performance of RT5 is insignificant. For the stock selection component, RT5's annualized return on investment is 0.14%, accounting for 13.81% of RT5's total stock selection, with a *t*-statistic of 2.56. This finding is consistent with our earlier result that RT5 might be better informed around EA days. For INST, the total performance over EA days is 0.11%, accounting for 9.80% of the total performance,

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with a significant *t*-statistic. The stock selection component shows a similar pattern. These findings indicate that institutional investors perform better on EA days than on all days.[27]

V. Robustness

A. Subperiods: Predictive Patterns for Different Years

Do the predictive patterns of retail order flows differ when market conditions change? Our 3.5-year sample period provides a good setting with different market conditions. The market return is -8.4% in 2016, 12.1% in 2017, -22.2% in 2018, and 21.2% in 2019. With two positive and negative returns, our sample period provides settings for both market-ups and-downs. To examine the predictive patterns of order flows for different years, we modify the Fama and MacBeth (1973) specification in equation (2) and interact the order flow variables with year dummies. The coefficients on the interaction terms provide information on whether the predictive pattern changes on each year.

[Insert Table IX about here]

From Table IX Panel A, the negative predictive pattern of order flow from RT1-RT4 for the next-day return, as observed in Table II, is robust for different years. Moreover, it is interesting to notice that the negative magnitudes are generally larger for 2016 and 2018, indicating that the negative predictive pattern is even larger in a downward market. For the largest retail investor, RT5, the positive predictive pattern is also larger during 2016 and 2018, indicating that their information advantage, if any, may also be stronger when the market is down. Similar patterns are observed for institutional investors.

[27] Previous literature (Kaniel et al., 2008; Barrot et al., 2016) argue that retail investors provide liquidity to the market and are compensated for their liquidity provision. We further decompose retail investors' performances from the perspective of liquidity provision, and provide the results in Appendix Table VII. Both small and large retail investors' performance are significantly negative if the trades demand liquidity. But if they provide liquidity, small retail investors' performance become insignificantly different from zero, and large retail investors are compensated for liquidity provision, which generally supports the hypothesis that liquidity provision is compensated.

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B. Subgroups: Predicting Patterns Across Firms with Different Characteristics

Previous studies show that stock returns can be significantly affected by firm and stock characteristics, such as size, EP ratio, and liquidity. Do the predictive patterns of retail order flows differ across firms with different characteristics? To answer this question, we modify the Fama and MacBeth (1973) specification in equation (2) and allow different coefficients for firms with different characteristics, by including interactions with characteristic dummies. Consider the size as an example. We first divide all firms on day d into three groups based on the previous month-end firm market capitalization. The dummy variable, $small\_size_{i,d-1}$, takes the value 1 if firm i belongs to the smallest 1/3 of firms and zero otherwise; $median_size_{i,d-1}$takes the value 1 if firm i belongs to the medium 1/3 of firms and zero otherwise; and $large\_size_{i,d-1}$ takes the value 1 if firm i belongs to the largest 1/3 of firms and zero otherwise. The coefficients provide information on whether the predictive pattern changes for firms of different sizes.

Panel B of Table IX reports the results. In the first three rows, we separate the firms based on their market capitalization. The negative predictive pattern of order flow from RT1-RT4 for the next-day return, as shown in Table II, is quite robust for firms with different sizes. However, it is interesting to notice that the magnitudes generally decrease from the smallest to the largest firms, indicating that the negative predictive pattern is the strongest for smaller firms. For large retail investors, RT5, the positive predictive pattern remains for small- and medium-sized firms, but not for large firms, indicating that their information about future returns, if any, might be concentrated in smaller firms. In comparison, order flows from institutions significantly predict next-day returns in all three rows, and more so for large firms, suggesting that their information about future returns, if any, might be more prominent for larger firms. When we separate firms by EP, turnover, and stock prices, we observe similar patterns. In other words, the predictive patterns in Table II are

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generally robust across firms with different characteristics, and the negative (positive) predictive power of smaller (larger) retail investors is stronger for small, low-EP, and higher-turnover firms, while the positive predictive power of institutional investors is stronger for large and high-EP firms.[28]

C. Different Filters

We consider three different sets of filters in the robustness check. In the main results, we discard stocks with fewer than 15 days of trading during the most recent month. In addition to this filter, Liu et al. (2019) also eliminate stocks that have become public within the past six months, stocks with less than 120 days of trading during the past 12 months, and the smallest 30% of all firms listed in China A-share market. We add all these additional filters and check the robustness of our results. In Panel C of Table IX, the order imbalance prediction directions are similar to those in Table II and the economic magnitude of institutions remains large. Thus, our main results are robust to the stricter filters from Liu et al. (2019).

Our second set of filters relates to leverage trading. Our trade-level data identify investors' margin buys, short sales, and collateral trades. Leveraged trading may differ from non-leverage trading. Each day, margin buys account for 10% of the trading volume, short sales account for 0.2%, and collateral trading accounts for 15% during our sample period. We exclude leverage trades and re-estimate equation (2). Panel D of Table IX show that the order imbalance prediction directions are similar to the results in Table II and the economic magnitudes are quantitatively similar. That is, our results are also robust to whether or not we include these leverage trades.

[28] We obtain a three-month sample from January 2019 to March 2019 with investors' gender and age information, and examine whether there are significant differences between male/female and young/old. The results are reported in Appendix Table VI. Across all gender-age groups, older male investors trade the most. For return prediction, the male investors across all ages significantly and negatively predict returns, especially for older males, while they are insignificant and different from zero for female retail investors. These patterns are mostly consistent with previous findings in Barber and Odean (2001).

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Finally, we consider the price limits. One institutional feature of the Chinese market is price limit restrictions. That is, investors can buy and sell stocks freely when the stock's price is within ±10% of the previous day's closing price. If the price moves out of the ±10% range, trading stops until the next day opens. Chen et al. (2019) focus on price limit days and find that large investors tend to trade differently on these days. Here, we examine whether our results hold when we exclude price limit days. We re-estimate equation (2) without the price limit days and present the results in Panel E of Table IX. Our results are robust to whether or not we include these price limit days.

D. Alternative Portfolio Forming Method

Our main results on return predictive patterns are based on the Fama and MacBeth (1973) regressions, which assume linear relations between future returns and order flow variables. In this section, we adopt an alternative portfolio approach and examine whether our results still hold. More specifically, we sort firms into five groups each day based on the previous day's order imbalance from a particular investor group, buying and selling 20% of stocks with the highest and lowest order imbalance measures for that particular investor group. We report the risk-adjusted returns (alphas) on this long-short strategy for one to 60 days, where we conduct risk adjustment using the Liu et al. (2019) three factor model.

From Panel F of Table IX, the one-day long-short portfolio alpha, using the previous day order imbalance from RT1, is -0.0042 and highly significant. From one week to 12 weeks, the cumulative alphas for the long-short portfolio decrease from -0.0089 to -0.0183, and they are all highly significant. That is, the cumulative alphas using *OibRT1* are consistently negative and significant, and there are no signs of reversal within 12 weeks, which echoes our earlier results in Tables II and V. Similar patterns exist for RT2 and RT3. For RT4, the one-day alpha is negative at -0.0007, but quickly becomes insignificant when we extend the holding horizon to one week,

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indicating that RT4 for horizons longer than one day. For the largest retail investors in RT5, the one-day alpha is 0.0017, which is positive and significant. The 12-week cumulative alpha is 0.0057, still positive and significant, confirming the results in Tables II and V that RT5 has both short- and long-term predictive power for future returns. The results for *OibINST* are similar to those for OibRT5.

V. Conclusion

Using comprehensive retail trading and holding data from 2016 to 2019, we divide tens of millions of retail investors into five groups based on their account sizes and examine their roles in the price discovery process in terms of return predictive power and the driving forces of these predictive patterns. Retail investors with account sizes of less than 3 mil CNY buy and sell stocks in the wrong direction. The prices of stocks they buy experience negative returns the next day, whereas those they sell experience positive returns. For retail investors with large account balances, trading predicts returns correctly. By tracing their differences in predicting future returns, we provide evidence that the negative predictive power of retail investors with smaller account sizes is mostly related to their order persistence, daily momentum trading, behavioral biases, and failures in processing earnings news. By contrast, the positive predictive power of large retail investors is mostly associated with order persistence, contrarian trading, trading against behavioral biases, and advantages in processing earnings news. Following BLLO (2009), we construct net buy and sell portfolios for each group of investors, and track their trading flows. This performance generates results consistent with the predictive patterns, in the sense that smaller retail investors have low stock selection skills and market timing skills, while large retail investors have better stock selection skills.

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Our study of the heterogeneous trading behavior of Chinese retail investors provides many unique insights into this large group of investors. In addition, the exchange acknowledges the heterogeneity of retail investors and focuses on adopting policies on investor education and suitability that restrict some kinds of trading for the smallest accounts. For example, retail investors are required to have at least 500k CNY holdings of stocks for at least 20 trading days to open a leverage trading account or to trade on a riskier Science and Technology Innovation Board (or STAR Market). These policies effectively exclude the smallest retail investors from leverage trading and trading on riskier start-ups, which could help protect them from even worse losses. Our study also raises many interesting questions. For example, why do retail investors dominate trading in the Chinese stock market? What role do institutional investors play? We leave these interesting and important questions to future research.

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Table I. Summary Statistics

This table reports summary statistics for trading and holdings by different investor groups. Our sample period is January 2016 to June 2019, and our sample firms are A-share stocks listed on a major stock exchange with at least 15 trading days during the previous month. Panel A reports the number of accounts, aggregate trading and holdings by different types of investors. Panel B reports the time series average of the cross-sectional distribution of stock characteristics. Panel C presents the time series average of the cross-sectional statistics of order imbalances for each investor group. Order imbalances (*Oib*) are computed as the buy share volume minus sell share volume divided by buy share volume plus sell share volume for each investor group, as specified in equation (1).

Panel A. Number of accounts, trading, and holdings by different types of investors

	RT1	RT2	RT3	RT4	RT5	INST	CORP
Account value	<100K	(100K,500K)	(500K,3M)	(3M,10M)	>10M CNY		
Number of Accounts (thousands)	31,410	15,282	5,827	735	235	40	47
Aggregate Trading Volume (Bil. CNY)	9	35	54	27	37	35	3
Aggregate Trading Volume (% of total)	5%	17%	27%	13%	19%	17%	2%
Aggregate Holdings Value (Bil CNY)	336	951	1,566	840	1,794	4,201	15,547
Aggregate Holdings Value (% of total)	1%	4%	6%	3%	7%	17%	62%

Panel B. Stock characteristics

	Mean	Std	P25	P50	P75
Market Capitalization (Billion CNY)	20.1	80.3	2.9	5.6	12.1
Earnings to Price Ratio	0.0075	0.0155	0.0018	0.0060	0.0122
Daily Stock Return	-0.01%	2.17%	-1.09%	-0.22%	0.77%
Daily Turnover (of tradable A shares)	2.45%	4.37%	0.61%	1.15%	2.40%

Panel C. Order imbalance in the cross section by different types of investors

	Mean	Std	AR1	Correlations OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	OibINST	OibCORP
OibRT1	-0.021	0.187	0.243	1						
OibRT2	-0.011	0.171	0.259	0.802	1					
OibRT3	-0.006	0.166	0.216	0.610	0.710	1				
OibRT4	0.002	0.250	0.059	0.194	0.244	0.256	1			
OibRT5	0.019	0.352	0.102	-0.151	-0.158	-0.163	-0.091	1		
OibINST	-0.011	0.455	0.205	-0.315	-0.365	-0.380	-0.263	-0.188	1	
OibCORP	-0.004	0.720	0.088	0.022	0.029	0.021	-0.007	-0.043	-0.044	1

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Table II. Predicting Next Day Stock Returns Using Order Imbalances from Different Investor Groups

This table reports the estimation results of whether trading activity by different investor groups can predict the cross-section of future stock returns. The coefficients are estimated from Fama-MacBeth (1973) regressions, as specified in equation (2). The dependent variable is the next-day stock return (*Ret*) and the independent variable is the previous day order imbalance *Oib*(-1). The control variables are the previous day return *Ret*(-1), previous week return *Ret*(-6,-2), previous month return *Ret*(-27,-7), previous month log market cap (*Size*), earnings to price ratio (*EP*), and monthly turnover (*Turnover*). The interquartile range for the relevant explanatory order imbalance is to compute the difference in predicted future returns for the interquartile range (*Interquartile return*). To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Dep.var		Ret	Ret	Ret	Ret	Ret	Ret
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	OibINST
Oib(-1)	Estimate	-0.0093***	-0.0091***	-0.0065***	-0.0009***	0.0012***	0.0016***
	[*t*-stat]	[-17.36]	[-17.10]	[-15.74]	[-6.95]	[10.48]	[18.06]
Ret(-1)	Estimate	-0.0027	-0.0091**	0.0006	0.0189***	0.0190***	0.0132**
	[*t*-stat]	[-0.61]	[-2.09]	[0.12]	[3.68]	[3.74]	[2.49]
Ret(-6,-2)	Estimate	-0.0149***	-0.0132***	-0.0124***	-0.0120***	-0.0115***	-0.0113***
	[*t*-stat]	[-8.17]	[-7.15]	[-6.66]	[-6.41]	[-6.17]	[-6.08]
Ret(-27,-7)	Estimate	-0.0039***	-0.0036***	-0.0034***	-0.0033***	-0.0032***	-0.0034***
	[*t*-stat]	[-4.35]	[-4.08]	[-3.84]	[-3.70]	[-3.61]	[-3.84]
Size	Estimate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	[*t*-stat]	[0.36]	[0.17]	[-0.16]	[-0.32]	[-0.18]	[-0.21]
EP	Estimate	0.0147***	0.0150***	0.0145***	0.0144***	0.0146***	0.0140***
	[*t*-stat]	[2.88]	[2.89]	[2.75]	[2.77]	[2.82]	[2.72]
Turnover	Estimate	-0.0007***	-0.0007***	-0.0007***	-0.0007***	-0.0007***	-0.0007***
	[*t*-stat]	[-3.50]	[-3.65]	[-3.64]	[-3.79]	[-3.79]	[-3.79]
Intercept	Estimate	-0.0012	-0.0006	0.0002	0.0005	0.0001	0.0002
	[*t*-stat]	[-0.47]	[-0.25]	[0.06]	[0.20]	[0.06]	[0.10]
Adj.R2		8.83%	8.68%	8.43%	8.10%	8.11%	8.25%
Interquartile oib		0.2222	0.1827	0.1678	0.2868	0.4536	0.6740
Interquartile return		-0.2062%	-0.1668%	-0.1089%	-0.0247%	0.0523%	0.1046%

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Table III. Predictive Patterns for Next Day Returns with Different Counterparties

This table reports the estimation results for the predictive patterns of next-day returns with different counterparties. For the counterparties, we put RT1 to RT4 together as one group and RT5 and INST as two separate groups. With the three groups of investors and two sides of trades (buy or sell), we divide the remaining trades into six bins: BBS, BSB, BSS, SBB, SBS, and SSB. The first letter indicates the trade direction of RT1-RT4, the second indicates the trade direction of RT5, and the third indicates INST. Panel A reports the coverage of each bin in the total sample. Panel B reports the coefficients estimated from the Fama-MacBeth (1973) regressions, as specified in equation (3). The control variables are the same as those in Table II and are not reported for abbreviations. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Distribution of counterparties

	RT1-4	RT5	INST	Proportions of Total Sample
BBS	Buy	Buy	Sell	17%
BSB	Buy	Sell	Buy	11%
BSS	Buy	Sell	Sell	21%
SBB	Sell	Buy	Buy	22%
SBS	Sell	Buy	Sell	13%
SSB	Sell	Sell	Buy	17%

Panel B. Predictive patterns with different counterparties

		I	II	III
Dep.var		Ret	Ret	Ret
$b1_{k,G}$		OibRT1-4	OibRT5	OibINST
Oib(-1)*BBS	Estimate	-0.0020***	-0.0007***	0.0021***
	[*t*-stat]	[-2.85]	[-3.21]	[11.39]
Oib(-1)*BSB	Estimate	-0.0041***	0.0004**	-0.0014***
	[*t*-stat]	[-3.24]	[2.26]	[-4.53]
Oib(-1)*BSS	Estimate	-0.0079***	0.0036***	0.0036***
	[*t*-stat]	[-12.10]	[13.91]	[15.11]
Oib(-1)*SBB	Estimate	-0.0145***	0.0032***	0.0019***
	[*t*-stat]	[-17.23]	[12.42]	[8.88]
Oib(-1)*SBS	Estimate	-0.0255***	0.0023***	-0.0013***
	[*t*-stat]	[-10.57]	[7.59]	[-3.33]
Oib(-1)*SSB	Estimate	-0.0093***	-0.0027***	0.0015***
	[*t*-stat]	[-13.20]	[-11.11]	[10.10]
Controls		Yes	Yes	Yes
Adj.R2		9.18%	8.41%	8.60%

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Table IV. Predictive Patterns for Next Day Return for Different Holding Horizons

This table reports the estimation results on the predictive patterns for the next day return for different holding horizons. We measure the holding days using the days-to-cover ratio (*DTCR*), defined as the total shares held by investor group *G* divided by daily shares traded on stock *i* for this investor group. Panel A reports the cross-sectional distribution of each investor group's holding horizons. Panel B reports the coefficients estimated from the Fama-MacBeth (1973) regressions, as specified in equation (4). We control the relative trading volume (*RVol*), defined as the daily trading volume divided by the average daily trading volume from previous 49 days. Other control variables are the same as those in Table II, and are not reported for brevity. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Distribution of holding horizon (days-to-cover-ratio)

	Mean	Std	P5	P25	P50	P75	P95
RT1	57	691	4	11	24	49	116
RT2	43	402	4	10	20	36	74
RT3	46	458	4	12	22	38	73
RT4	64	833	4	13	26	48	114
RT5	251	4,479	4	17	41	95	420
INST	199	3,750	3	12	31	72	265

Panel B. Predictive patterns for different holding horizons

Dep.var		Ret OibRT1	Ret OibRT2	Ret OibRT3	Ret OibRT4	Ret OibRT5	Ret OibINST
Oib(-1)*(0,10]Days	Estimate	-0.0171***	-0.0191***	-0.0172***	-0.0023***	0.0056***	0.0017***
	[*t*-stat]	[-29.84]	[-28.77]	[-23.05]	[-4.28]	[13.68]	[12.51]
Oib(-1)*(10,20]Days	Estimate	-0.0086***	-0.0078***	-0.0054***	-0.0006**	0.0015***	0.0012***
	[*t*-stat]	[-21.46]	[-19.74]	[-12.75]	[-2.43]	[7.07]	[9.62]
Oib(-1)*(20,60]Days	Estimate	-0.0064***	-0.0052***	-0.0033***	-0.0004***	0.0007***	0.0013***
	[*t*-stat]	[-18.24]	[-15.42]	[-11.82]	[-2.81]	[6.37]	[11.49]
Oib(-1)*Above60Days	Estimate	-0.0074***	-0.0007	-0.0086*	0.0008	0.0002	0.0008***
	[*t*-stat]	[-6.26]	[-0.12]	[-1.72]	[0.35]	[1.52]	[7.80]
Rvol(-1)	Estimate	-0.0007***	-0.0008***	-0.0011***	-0.0014***	-0.0013***	-0.0013***
	[*t*-stat]	[-5.91]	[-6.97]	[-9.03]	[-11.31]	[-10.82]	[-10.88]
Controls		Yes	Yes	Yes	Yes	Yes	Yes
Adj.R2		10.03%	9.94%	9.72%	9.24%	9.31%	9.26%

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Table V. Predict Returns over the Next 12 Weeks

This table reports the estimation results of whether trading activity by different investor groups can predict the cross-section of returns over the next 12 weeks. We report the coefficient estimates from the Fama-MacBeth (1973) regressions specified in equation (5). The main independent variables are the previous day's order imbalance, *Oib(-1)*. The control variables are the same as those in Table II, and are not reported for brevity. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

*w*th weeks		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	OibINST
1	Estimate	-0.0226***	-0.0220***	-0.0144***	-0.0019***	0.0027***	0.0044***
	[*t*-stat]	[-18.84]	[-16.98]	[-13.43]	[-6.70]	[10.17]	[17.98]
2	Estimate	-0.0065***	-0.0060***	-0.0037***	0.0001	0.0010***	0.0012***
	[*t*-stat]	[-9.83]	[-8.56]	[-5.89]	[0.50]	[6.01]	[5.64]
3	Estimate	-0.0038***	-0.0031***	-0.0015**	0.0001	0.0007***	0.0007***
	[*t*-stat]	[-5.37]	[-4.09]	[-2.31]	[0.45]	[3.77]	[3.17]
4	Estimate	-0.0024***	-0.0021**	-0.0012	-0.0001	0.0007***	0.0005**
	[*t*-stat]	[-3.46]	[-2.57]	[-1.63]	[-0.37]	[3.94]	[2.09]
5	Estimate	-0.0014***	-0.0014**	-0.0011**	-0.0005**	0.0004**	0.0002
	[*t*-stat]	[-2.69]	[-2.44]	[-2.07]	[-2.00]	[2.25]	[1.39]
6	Estimate	-0.0029***	-0.0024***	-0.0016***	-0.0003	0.0001	0.0005***
	[*t*-stat]	[-4.69]	[-3.55]	[-2.71]	[-1.22]	[0.72]	[2.66]
7	Estimate	-0.0027***	-0.0025***	-0.0018***	-0.0002	0.0001	0.0007***
	[*t*-stat]	[-5.02]	[-4.43]	[-3.65]	[-0.68]	[0.39]	[4.25]
8	Estimate	-0.0015**	-0.0010	-0.0007	-0.0002	0.0003**	0.0004**
	[*t*-stat]	[-2.57]	[-1.52]	[-1.08]	[-0.73]	[2.04]	[2.39]
9	Estimate	-0.0010*	-0.0005	-0.0004	0.0002	0.0004**	0.0001
	[*t*-stat]	[-1.94]	[-1.00]	[-0.73]	[1.03]	[2.43]	[0.43]
10	Estimate	-0.0007	-0.0004	-0.0004	0.0002	-0.0001	0.0002
	[*t*-stat]	[-1.22]	[-0.60]	[-0.81]	[0.94]	[-0.44]	[0.90]
11	Estimate	-0.0006	-0.0007	-0.0001	0.0000	-0.0002	0.0002
	[*t*-stat]	[-1.02]	[-1.01]	[-0.12]	[-0.04]	[-1.10]	[1.10]
12	Estimate	-0.0005	-0.0001	0.0005	0.0001	0.0001	0.0000
	[*t*-stat]	[-0.90]	[-0.12]	[0.76]	[0.41]	[0.39]	[0.05]

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Table VI. Two Stage Decomposition for Understanding the Predictive Patterns of Retail Order Flows

This table reports the estimation results of the decomposition of the predictive power of different investor groups' order imbalances for the cross-section of future stock returns. We estimate two-stage Fama-MacBeth (1973) regressions. Panel A reports the first-stage estimation results, where the order imbalance measures are decomposed into five components as specified in equation (A1). Variable *Overconf*(-1) is measured as the corresponding investor group's average turnover on the stock from the previous 20 days as a proxy for each investor's overconfidence. The variable *Gamble*(-1) is the maximum daily return from the previous 20 days as a proxy for gambling preference. Panel B reports the second-stage decomposition of the order imbalance's predictive power, as specified in equations (A2) and (A3). As an independent variable, the variable *Oib*(-1,*Persistence*) is estimated in the first stage using the past order imbalance and reflects price pressure. The variable *Oib*(-1,*Liquidity*) is estimated in the first stage using past returns over different horizons, and is connected to the liquidity provision or liquidity demand hypothesis. The variable *Oib*(-1,*Overconf*) is estimated in the first stage, reflecting overconfidence. The variable *Oib*(-1,*Gamble*) is estimated in the first stage using the maximum daily returns from the previous 20 days and reflects a preference for gambling. The residual part of the previous day order imbalance from the first-stage estimation is denoted "other," which can be attributed to private information about future returns. The control variables are the same as those in Table II, and are not reported for brevity. For each regression, we also report the difference in predicted day-ahead returns for observations at the two ends of the interquartile range (*Interquartile return*) in Panel B. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

Panel A. First stage of projecting the order imbalance on persistence, past returns, overconfidence, and gambling proxies

Dep.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib(-1)	Estimate	0.1870***	0.1967***	0.1711***	0.0499***	0.1036***
	[*t*-stat]	[24.24]	[32.92]	[35.03]	[15.03]	[27.85]
Ret(-1)	Estimate	0.5159***	0.7269***	0.4482***	-0.2205***	-1.2968***
	[*t*-stat]	[9.17]	[16.36]	[14.52]	[-7.57]	[-28.84]
Ret(-6,-2)	Estimate	-0.4214***	-0.2196***	-0.1063***	-0.0804***	-0.0485***
	[*t*-stat]	[-19.83]	[-15.14]	[-7.71]	[-4.22]	[-2.74]
Ret(-27,-7)	Estimate	-0.0350***	-0.0196***	-0.0235***	-0.0399***	-0.0203***
	[*t*-stat]	[-6.49]	[-4.89]	[-6.72]	[-8.36]	[-3.74]
Overconf(-1)	Estimate	0.0894***	0.0611***	0.0657***	0.0418***	-0.0881***
	[*t*-stat]	[4.53]	[4.57]	[7.28]	[3.64]	[-8.80]
Gamble(-1)	Estimate	0.0330	0.0784***	0.1731***	0.2423***	-0.0671**
	[*t*-stat]	[1.49]	[5.06]	[11.69]	[11.90]	[-2.53]
Intercept	Estimate	-0.0214***	-0.0120***	-0.0115***	-0.0078***	0.0231***

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	[*t*-stat]	[-5.70]	[-5.09]	[-7.63]	[-4.00]	[9.41]
Adj.R2		7.08%	5.60%	3.89%	0.73%	2.00%

Panel B. Second stage decomposition of order imbalance's predictive power

Dep.var		Ret	Ret	Ret	Ret	Ret
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib(-1,Persistence)	Estimate	-0.0301***	-0.0276***	-0.0205***	-0.0086***	0.0060***
	[*t*-stat]	[-16.73]	[-15.39]	[-12.22]	[-4.37]	[7.87]
Oib(-1,Liquidity)	Estimate	-0.0135***	-0.0205***	-0.0261***	0.0140	0.0042*
	[*t*-stat]	[-4.48]	[-6.18]	[-4.69]	[0.69]	[1.69]
Oib(-1,Overconf)	Estimate	-0.1006***	-0.2158***	-0.2413***	-0.3730***	0.1134***
	[*t*-stat]	[-4.00]	[-5.97]	[-6.05]	[-5.85]	[5.90]
Oib(-1,Gamble)	Estimate	-0.2877***	-0.0981***	-0.0474***	-0.0389***	0.1918***
	[*t*-stat]	[-4.89]	[-4.08]	[-4.35]	[-4.98]	[6.35]
Oib(-1,Other)	Estimate	-0.0086***	-0.0084***	-0.0060***	-0.0008***	0.0011***
	[*t*-stat]	[-15.54]	[-15.51]	[-14.16]	[-6.19]	[9.59]
Adj.R2		10.50%	10.35%	10.07%	9.59%	9.52%
Interquartile return						
Oib(-1,Persistence)		-0.1177%	-0.0962%	-0.0591%	-0.0125%	0.0281%
Oib(-1,Liquidity)		-0.0290%	-0.0351%	-0.0261%	0.0091%	0.0097%
Oib(-1,Overconf)		-0.0337%	-0.0475%	-0.0484%	-0.0428%	0.0257%
Oib(-1,Gamble)		-0.0314%	-0.0254%	-0.0271%	-0.0312%	0.0425%
Oib(-1,Other)		-0.1778%	-0.1493%	-0.1008%	-0.0233%	0.0490%

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Table VII. A Closer Look at the Relation between Investor Order Flows and Earnings News

This table reports the estimation results on the relation between different investor groups' order flows and public earnings announcements. Panel A reports whether investor order flow can predict earnings surprises, as specified in equation (6). The dependent variable is the cumulative abnormal return from day *d*-1 to day *d*, *CAR*[-1,0], and the independent variable is order imbalance from day *d*-2, *Oib*(-2). Panel B reports whether trades from different retail groups can process contemporaneous news, as specified in equation (7). The dependent variable is the order imbalance *Oib*(0), and the independent variable is the cumulative abnormal return from day *d*-1 to day *d*, *CAR*[-1,0]. Panel C reports the effect of earnings news days on the return predictability of different investor group trades, as specified in equation (8). The dependent variable is the return on day *d*, and the independent variables include the previous day's order imbalance *Oib*(-1), the news dummy *Event*(-1), and the interaction terms *Oib*(-1)***Event*(-1). The *Event*(-1) dummy equals 1 if there is an earnings announcement for that firm day and zero otherwise. The other control variables are the same as those in Table II, and these coefficients are not reported. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Investor order flow predicting future earnings announcement news events

Dep.var		CAR[-1,0]	CAR[-1,0]	CAR[-1,0]	CAR[-1,0]	CAR[-1,0]
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib(-2)	Estimate	-0.0251***	-0.0234***	-0.0166***	-0.0003	0.0023***
	[*t*-stat]	[-8.12]	[-5.81]	[-4.56]	[-0.50]	[3.67]
Adj.R2		6.33%	5.98%	5.57%	5.19%	5.15%

Panel B. Investor order flow regressed on contemporaneous earnings announcement news events

Dep.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
CAR[-1,0]	Estimate	-1.9225***	-1.8291***	-1.4349***	-0.8781***	0.1583**
	[*t*-stat]	[-22.65]	[-40.09]	[-14.53]	[-5.40]	[2.12]
Adj.R2		13.66%	14.60%	10.14%	1.85%	0.60%

Panel C. Return predictive power of investor order flow interacted with earnings announcement news events

Dep.var		Ret	Ret	Ret	Ret	Ret
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib(-1)	Estimate	-0.0079***	-0.0070***	-0.0038***	-0.0008*	0.0005**
	[*t*-stat]	[-9.33]	[-7.67]	[-5.55]	[-1.93]	[3.08]
Oib(-1)*Event(-1)	Estimate	-0.0080***	-0.0093***	-0.0071**	-0.0007	0.0014**

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	[*t*-stat]	[-4.22]	[-3.90]	[-2.52]	[-1.54]	[2.07]
Event(-1)	Estimate	0.0011***	0.0008****	0.0006***	0.0005**	0.0005**
	[*t*-stat]	[3.94]	[2.89]	[2.80]	[2.55]	[2.40]
Adj.R2		7.36%	7.16%	6.82%	6.38%	6.35%

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Table VIII. Annual Performance Using Tracking Portfolios of Retail Trading, from Stock Selection and Market Timing

This table reports the annual performance for different retail investor groups using Barber, Lee, Liu, and Odean's (2009) approach between January 2016 and June 2019. The net buy and sell portfolios are constructed each day, as specified in equation (A4), and the average construction prices are defined in equation (A6). The daily portfolio's total performance of holding *n* days is the portfolio cumulative excess return minus the trading cost, as specified in equation (A5), and we choose *n* equals to 140 days following Barber et al. (2009). We then annualize the daily portfolio's total performance by timing the 240 trading days on each year and dividing by aggregate holding value in Table I Panel A to present the investor group's annual performance in percentage. The total performance could be decomposed into stock selection, market timing, and trading cost, as specified in equation (A7)-(A10). Panel A presents the total annual performance. Panel B reports annual performance conditional on the momentum and contrarian. If the investor group *G*'s order imbalance times previous day return, $Oib(i,d,G) * Ret(i,d-1) > 0$, we classify the stock *i* on day *d* for investor group *G* to be momentum; if $Oib(i,d,G) * Ret(i,d-1) \leq 0$,we classify the stock *i* on day *d* for investor group *G* to be contrarian. Panel C reports the annual performance of the earnings announcements. Earnings announcement events include the stock on announcement day *d* and the previous day *d*-1. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Annual trading performance using tracking portfolios

		Total	Stock Selection	Market Timing	Trading Cost
RT1	Performance	-5.61%***	-4.03%***	-0.19%	-1.38%***
	[*t*-stat]	[-10.29]	[-8.39]	[-0.56]	[-23.47]
RT2	Performance	-4.61%***	-2.55%***	-0.27%	-1.80%***
	[*t*-stat]	[-8.96]	[-6.21]	[-0.93]	[-23.07]
RT3	Performance	-3.60%***	-1.60%***	-0.29%*	-1.72%***
	[*t*-stat]	[-9.49]	[-5.51]	[-1.71]	[-20.89]
RT4	Performance	-2.80%***	-0.90%***	-0.33%*	-1.58%***
	[*t*-stat]	[-8.51]	[-3.98]	[-1.78]	[-19.53]
RT5	Performance	-0.29%	1.05%***	-0.30%	-1.03%***
	[*t*-stat]	[-0.92]	[5.27]	[-1.39]	[-18.84]
INST	Performance	1.15%***	1.38%***	0.19%	-0.41%***
	[*t*-stat]	[3.57]	[4.85]	[0.93]	[-20.49]

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Panel B. Annual trading performance using tracking portfolios: separate momentum and contrarian trading

		Total		StockSelection		MarketTiming		TradingCost	
		Contrarian	Momentum	Contrarian	Momentum	Contrarian	Momentum	Contrarian	Momentum
RT1	Performance	-1.04%**	-4.57%***	0.07%	-4.10%***	-0.41%	0.22%	-0.70%***	-0.68%***
	[*t*-stat]	[-2.57]	[-8.47]	[0.25]	[-8.51]	[-1.39]	[1.03]	[-23.33]	[-21.29]
RT2	Performance	-0.95%**	-3.67%***	0.43%	-2.97%***	-0.46%*	0.19%	-0.91%***	-0.89%***
	[*t*-stat]	[-2.29]	[-7.88]	[1.46]	[-7.63]	[-1.79]	[1.12]	[-23.61]	[-21.15]
RT3	Performance	-1.14%***	-2.47%***	0.17%	-1.77%***	-0.42%**	0.14%	-0.88%***	-0.83%***
	[*t*-stat]	[-4.04]	[-8.00]	[0.85]	[-7.48]	[-2.47]	[1.02]	[-21.06]	[-19.65]
RT4	Performance	-0.95%***	-1.85%***	0.33%**	-1.23%***	-0.44%***	0.12%	-0.84%***	-0.74%***
	[*t*-stat]	[-4.09]	[-7.23]	[2.01]	[-6.66]	[-3.32]	[0.65]	[-19.59]	[-18.95]
RT5	Performance	1.32%***	-1.61%***	2.45%***	-1.40%***	-0.56%***	0.26%	-0.57%***	-0.46%***
	[*t*-stat]	[3.76]	[-4.13]	[7.96]	[-5.32]	[-3.03]	[1.18]	[-18.58]	[-18.66]
INST	Performance	0.64%***	0.51%**	0.89%***	0.49%***	-0.05%	0.23%	-0.19%***	-0.22%***
	[*t*-stat]	[4.49]	[2.04]	[6.71]	[3.04]	[-0.62]	[1.32]	[-18.23]	[-19.52]

Panel C. Annual trading performance using tracking portfolios: around earnings announcements

		Total		Stock Selection		Market Timing		Trading Cost	
		EA	EA/all days	EA	EA/all days	EA	EA/all days	EA	EA/all days
RT1	Performance	-0.39%***	6.89%	-0.36%***	8.96%	0.02%	-8.63%	-0.04%***	3.06%
	[*t*-stat]	[-3.68]		[-4.14]		[0.57]		[-5.91]	
RT2	Performance	-0.36%***	7.71%	-0.31%***	12.27%	0.01%	-4.85%	-0.06%***	3.12%
	[*t*-stat]	[-4.08]		[-3.51]		[0.47]		[-5.90]	
RT3	Performance	-0.24%***	6.61%	-0.18%***	11.46%	0.00%	0.03%	-0.05%***	3.18%
	[*t*-stat]	[-4.24]		[-2.95]		[-0.01]		[-5.86]	
RT4	Performance	-0.14%***	5.02%	-0.07%	8.36%	-0.01%	4.32%	-0.05%***	3.26%
	[*t*-stat]	[-2.70]		[-1.10]		[-1.32]		[-5.86]	
RT5	Performance	0.10%	-33.28%	0.14%**	13.81%	-0.01%	4.77%	-0.03%***	3.37%
	[*t*-stat]	[0.91]		[2.56]		[-1.30]		[-5.87]	
INST	Performance	0.11%*	9.80%	0.12%**	9.01%	0.00%	1.92%	-0.01%***	3.56%
	[*t*-stat]	[1.81]		[2.01]		[0.26]		[-5.86]	

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Table IX. Robustness

This table reports the robustness results. The sample period covers January 2016 to June 2019, and the sample firms are A-share stocks merged with the proprietary data and have at least 15 trading days in the previous month. Panel A reports the return prediction on each year. Panel B reports the return prediction by different stock characteristics. Panel C reports return predictions by including all filters in Liu, Stambaugh and Yuan (2019). Panel D reports return predictions by excluding leveraged trading, which includes margin buys, short sales, and collateral trading. Panel E reports return predictions by excluding days that hit the price limits (+10% and -10%). Panel F reports the alternative method using Liu, Stambaugh, and Yuan's (2019) three-factor adjusted alphas for long-short portfolios. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Cross sectional return predictions by different years

Year	Market Return	Dep.var Oib.var		Ret OibRT1	Ret OibRT2	Ret OibRT3	Ret OibRT4	Ret OibRT5	Ret OibINST
2016	-8.4%	Oib(-1)*Y2016	Estimate	-0.0027***	-0.0025***	-0.0020***	-0.0004***	0.0004***	0.0003***
			[*t*-stat]	[-3.46]	[-3.41]	[-3.45]	[-3.37]	[3.43]	[3.26]
2017	12.1%	Oib(-1)*Y2017	Estimate	-0.0023***	-0.0022***	-0.0015***	-0.0002***	0.0003***	0.0004***
			[*t*-stat]	[-3.83]	[-3.70]	[-3.71]	[-3.11]	[3.39]	[3.87]
2018	-22.2%	Oib(-1)*Y2018	Estimate	-0.0026***	-0.0027***	-0.0018***	-0.0002**	0.0002***	0.0006***
			[*t*-stat]	[-3.75]	[-3.80]	[-3.70]	[-2.17]	[3.45]	[3.81]
2019	21.2%	Oib(-1)*Y2019	Estimate	-0.0018**	-0.0018**	-0.0013**	-0.0001*	0.0003**	0.0003**
Jan-Jun			[*t*-stat]	[-2.25]	[-2.26]	[-2.29]	[-1.84]	[2.13]	[2.35]
		Control		Yes	Yes	Yes	Yes	Yes	Yes
		InterquartilereturnY2016		-0.0569%	-0.0434%	-0.0289%	-0.0088%	0.0136%	0.0248%
		InterquartilereturnY2017		-0.0478%	-0.0377%	-0.0240%	-0.0053%	0.0127%	0.0255%
		InterquartilereturnY2018		-0.0603%	-0.0527%	-0.0341%	-0.0060%	0.0129%	0.0375%
		InterquartilereturnY2019		-0.0414%	-0.0339%	-0.0215%	-0.0033%	0.0123%	0.0160%

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Panel B. Cross-sectional return predictions by different stock characteristics

Oib.var	OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	OibINST
Oib(-1)*Small_size	-0.0119***	-0.0131***	-0.0090***	-0.0006***	0.0020***	0.0014***
Oib(-1)*Median_size	-0.0096***	-0.0093***	-0.0066***	-0.0011***	0.0009***	0.0015***
Oib(-1)*Large_size	-0.0067***	-0.0061***	-0.0044***	-0.0012***	0.0001	0.0021***
Oib(-1)*Low_EP	-0.0122***	-0.0131***	-0.0095***	-0.0010***	0.0020***	0.0015***
Oib(-1)*Median_EP	-0.0097***	-0.0098***	-0.0069***	-0.0009***	0.0012***	0.0015***
Oib(-1)*High_EP	-0.0062***	-0.0056***	-0.0039***	-0.0007***	0.0001	0.0017***
Oib(-1)*Low_turnover	-0.0061***	-0.0057***	-0.0041***	-0.0007***	0.0003***	0.0013***
Oib(-1)*Median_turnover	-0.0091***	-0.0092***	-0.0066***	-0.0010***	0.0011***	0.0014***
Oib(-1)*High_turnover	-0.0159***	-0.0176***	-0.0128***	-0.0009***	0.0026***	0.0019***
Oib(-1)*Low_price	-0.0088***	-0.0086***	-0.0067***	-0.0012***	0.0009***	0.0014***
Oib(-1)*Median_price	-0.0098***	-0.0095***	-0.0068***	-0.0007***	0.0013***	0.0016***
Oib(-1)*High_price	-0.0094***	-0.0094***	-0.0060***	-0.0007***	0.0014***	0.0016***

Panel C. Cross-sectional return predictions, including all filters from Liu Stambaugh and Yuan (2019)

Dep.var		Ret	Ret	Ret	Ret	Ret	Ret
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	OibINST
Oib(-1)	Estimate	-0.0075***	-0.0071***	-0.0052***	-0.0011***	0.0004***	0.0017***
	[*t*-stat]	[-21.47]	[-19.65]	[-16.85]	[-8.09]	[5.07]	[16.98]
	Interquartile return	-0.17%	-0.14%	-0.09%	-0.03%	0.02%	0.11%

Panel D. Cross-sectional return predictions, excluding leveraged trading

Dep.var		Ret	Ret	Ret	Ret	Ret	Ret
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	OibINST
Oib(-1)	Estimate	-0.0092***	-0.0088***	-0.0056***	-0.0003***	0.0008***	0.0016***
	[*t*-stat]	[-17.26]	[-17.16]	[-15.54]	[-3.37]	[10.09]	[18.28]
	Interquartile return	-0.20%	-0.16%	-0.11%	-0.01%	0.05%	0.11%

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Panel E. Cross-sectional return predictions, excluding days that hitting the price limits (+10%, and -10%)

Dep.var		Ret	Ret	Ret	Ret	Ret	Ret
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	OibINST
Oib(-1)	Estimate	-0.0087***	-0.0087***	-0.0064***	-0.0010***	0.0009***	0.0017***
	[*t*-stat]	[-19.60]	[-19.57]	[-17.95]	[-8.38]	[10.41]	[18.10]
	Interquartile return	-0.19%	-0.16%	-0.11%	-0.03%	0.04%	0.11%

Panel F. Risk adjusted alphas for long-short portfolios over the next 12 weeks

	OibRT1		OibRT2		OibRT3		OibRT4		OibRT5		OibINST	
	Alpha	*t*-stat	Alpha	*t*-stat	Alpha	*t*-stat	Alpha	*t*-stat	Alpha	*t*-stat	Alpha	*t*-stat
1 day	-0.0042***	-20.55	-0.0036***	-17.53	-0.0027***	-15.13	-0.0007***	-6.71	0.0017***	12.71	0.0025***	19.28
1 week	-0.0089***	-15.34	-0.0068***	-12.97	-0.0038***	-9.22	-0.0004	-1.63	0.0034***	11.29	0.0056***	13.67
2 weeks	-0.0115***	-11.96	-0.0091***	-10.50	-0.0048***	-6.84	-0.0001	-0.16	0.0046***	9.08	0.0075***	11.53
3 weeks	-0.0130***	-10.52	-0.0103***	-9.37	-0.0051***	-5.68	0.0001	0.26	0.0054***	8.71	0.0086***	9.73
4 weeks	-0.0130***	-8.64	-0.0104***	-7.94	-0.0052***	-4.94	0.0003	0.81	0.0056***	7.87	0.0098***	9.19
5 weeks	-0.0130***	-7.45	-0.0099***	-6.53	-0.0050***	-4.01	0.0006	1.13	0.0062***	7.51	0.0098***	8.39
6 weeks	-0.0148***	-7.84	-0.0110***	-6.49	-0.0056***	-3.73	0.0005	0.98	0.0064***	7.29	0.0103***	7.99
7 weeks	-0.0173***	-8.76	-0.0130***	-6.74	-0.0068***	-3.61	0.0001	0.49	0.0064***	7.25	0.0119***	8.29
8 weeks	-0.0183***	-8.43	-0.0136***	-6.37	-0.0069***	-3.36	0.0001	0.50	0.0063***	6.63	0.0128***	6.95
9 weeks	-0.0191***	-7.62	-0.0142***	-5.59	-0.0070***	-2.97	0.0003	0.64	0.0068***	7.45	0.0135***	6.18
10 weeks	-0.0187***	-6.98	-0.0136***	-5.27	-0.0063***	-2.98	0.0005	0.87	0.0065***	7.68	0.0133***	6.38
11 weeks	-0.0187***	-6.24	-0.0141***	-4.66	-0.0070***	-2.77	-0.0005	0.27	0.0061***	6.83	0.0138***	6.00
12 weeks	-0.0183***	-5.75	-0.0139***	-4.52	-0.0072***	-2.89	-0.0010	0.03	0.0057***	5.78	0.0137***	6.03

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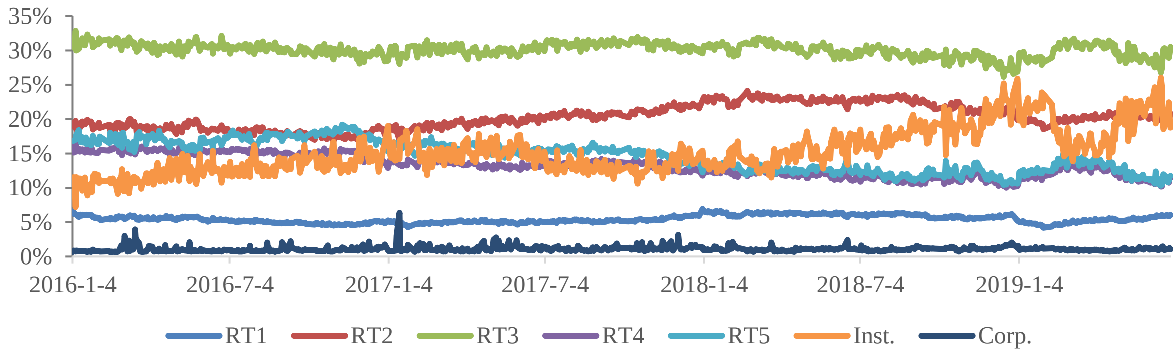
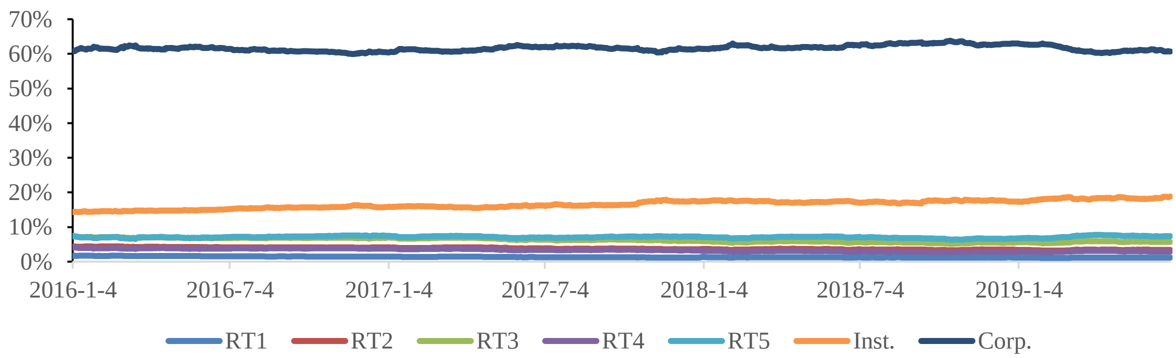
Figure I. Different Investor Type Order Flows between Jan 2016 and Jun 2019
The figures report the time-series plot of the different types of investor trading activity from January 2016 to June 2019. Our sample firms are A-share stocks listed on a major stock exchange with at least 15 trading days during the previous month. Panel A presents time-series plot of the volume percentage for each investor type, reported as cross-sectional mean. Panel B shows the time-series plot of shares held by investor type, aggregated at market level. Panel C presents the time-series plot of each investor type's trading volume, scaled by the shares held by that investor type, and reported as the cross-sectional mean.

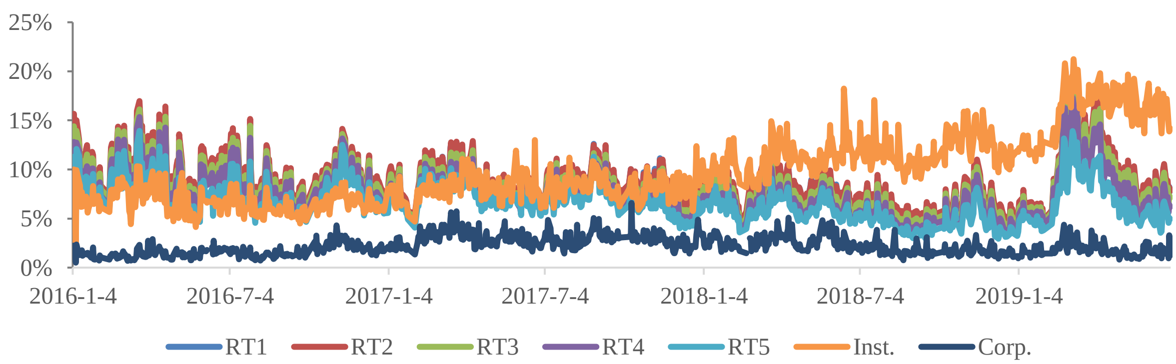
Panel A. Share volume (%), cross-sectional mean for different investor types



Panel B. Shares held (%), aggregated at market level for different investor types



Panel C. Share volume scaled by shares held (%), cross-sectional mean for different investor types



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Internet Appendix

Appendix A. Boehmer et al. (2021) Two-stage decomposition

Following Boehmer et al. (2021), we estimate the first stage of the two-stage decomposition. For each day *d*, we estimate the cross-sectional specification: [1]

$$Oib_{i,d} = h0_d + h1_d Oib_{i,d-1} + h2'_d Ret_{i,d-1} + h3_d Overconf_{i,d-1} + h4_d Gamble_{i,d-1} + u8_{i,d}. \quad \text{(A1)}$$

After we obtain the time-series of coefficients, $\{\widehat{h0}_d, \widehat{h1}_d, \widehat{h2}'_d, \widehat{h3}_d, \widehat{h4}_d\}$, we conduct statistical inference using the time-series means of the above coefficients, $\{\widehat{h0}, \widehat{h1}, \widehat{h2}', \widehat{h3}, \widehat{h4}\}$, and standard errors, which are adjusted using Newey and West (1987) method with the optimal bandwidth chosen by using Newey and West (1994) approach, to understand how each of the four hypotheses contributes to retail order flows. The first-stage estimation naturally decomposes $Oib_{i,d}$ into five components:

$$Oib_{i,d} = \widehat{Oib}_{i,d}^{persistence} + \widehat{Oib}_{i,d}^{liquidity} + \widehat{Oib}_{i,d}^{overconf} + \widehat{Oib}_{i,d}^{gamble} + \widehat{Oib}_{i,d}^{other}, \quad \text{(A2)}$$

With $\widehat{Oib}_{i,d}^{persistence} = \widehat{h1} Oib_{i,d}$, $\widehat{Oib}_{i,d}^{liquidity} = \widehat{h2}' Ret_{i,d-1}$, $\widehat{Oib}_{i,d}^{overconf} = \widehat{h3} Overconf_{i,d-1}$, $\widehat{Oib}_{i,d}^{gamble} = \widehat{h4} Gamble_{i,d-1}$, and $\widehat{Oib}_{i,d}^{other} = Oib_{i,d} - \widehat{Oib}_{i,d}^{persistence} - \widehat{Oib}_{i,d}^{liquidity} - \widehat{Oib}_{i,d}^{overconf} - \widehat{Oib}_{i,d}^{gamble}$. The "other" component is the residual component, which potentially contains other relevant information about future returns.

For the second stage of the decomposition, we relate future returns to the five components of order flow by the following specification using Fama and MacBeth's (1973) methodology:

$$Ret_{i,d+1} = m0_{d+1} + m1_{d+1}\widehat{Oib}_{i,d}^{persistence} + m2_{d+1}\widehat{Oib}_{i,d}^{liquidity} + m3_{d+1}\widehat{Oib}_{i,d}^{overconf} + m4_{d+1}\widehat{Oib}_{i,d}^{gamble} + m5_{d+1}\widehat{Oib}_{i,d}^{other} + m6'_{d+1} Controls_{i,d} + u9_{i,d+1}. \quad \text{(A3)}$$

The coefficient estimates in equation (A3) show how each component of order flows contributes to the predictive power of order flows for future stock returns. According to Boehmer et al. (2021), the advantage of the two-stage decomposition approach is that it includes various components of $Oib_{i,d}$ from the alternative hypotheses in a unified and internally consistent empirical framework. One caveat of this approach is that it is necessary to make empirical assumptions when choosing proxies for different hypotheses. Although these assumptions seem reasonable, we must be cautious as the results still depend on the validity of the empirical assumptions.

Appendix B. Barber et al. (2009) Methods for Trading Performances

With access to all trading records from the Taiwan Stock Exchange, Barber, Lee, Liu, and Odean (2009, BLLO hereafter) designs a simple and intuitive method to compute the trading performances of different types of investors in four steps. First, BLLO separate all investors into "individuals", "corporations", "dealers", "foreigners", and "mutual funds", using identities provided by the exchange. Second, within each investor group, BLLO compute the aggregate daily net buy and net sell positions and create daily matching trading portfolios, the "net buy" and "net sell" portfolios. Specifically, the net buy volumes and sell volumes for stock *i* on day *d* of group G are summed over investor *j* in group *G*:

$$NetBuyVol_{i,d,G} = max\left(\sum_{j\in G} BuyVol_{i,d,j} - \sum_{j\in G} SellVol_{i,d,j}, 0\right), \quad \text{(A4)}$$

[1] All estimations in this study are estimated within each investor group. Beginning in this section, we omit the subscripts G to make the formula more readable.

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$$NetSellVol_{i,d,G} = max\left(\sum\nolimits_{j\in G} SellVol_{i,d,j} - \sum\nolimits_{j\in G} BuyVol_{i,d,j}, 0\right),$$

where the $Buyvol_{i,d,j}$ and $Sellvol_{i,d,j}$ are defined in equation (1). If net buy volume for stock *i* on day *d* in investor group *G* is positive, it belongs to the net buy portfolio, $NetBuy_{d,G}$; if the net buy volume is negative (thus the net sell volume is positive), it belongs to the net sell portfolio, $NetSell_{d,G}$. Third, BLLO track these "net buy" and "net sell" portfolios over a holding horizon of *n* days, and compute cumulative cash flows and returns from this tracking strategy, and treat them as proxies for trading performances of different investor groups. Specifically, BLLO computes the total performance of the net buy and net sell portfolios as:

$$Total_{d,G} = \sum\nolimits_{i\in NetBuy_{d,G}} NetBuyVol_{i,G,d} \times AvgBuyPrice_{i,G,d} \times (ret_{i,d+1,d+n} - rf_{d+1,d+n}) - \sum\nolimits_{i\in NetSell_{d,G}} NetSellVol_{i,G,d} \times AvgSellPrice_{i,G,d} \times \left(ret_{i,d+1,d+n} - rf_{d+1,d+n}\right) - TrdCost_{d,G}. \tag{A5}$$

The tracking net buy and sell portfolios are rebalanced once a day, so we follow BLLO and use the average prices, defined as

$$AvgBuyPrice_{i,G,d} = \frac{\max\left(\sum_{j\in G} BuyVolCNY_{i,d,j} - \sum_{j\in G} SellVolCNY_{i,d,j}, 0\right)}{NetBuyVol_{i,d,G}}, \tag{A6}$$

$$AvgSellPrice_{i,G,d} = \frac{\max\left(\sum_{j\in G} SellvolCNY_{i,d,j} - \sum_{j\in G} BuyvolCNY_{i,d,j}, 0\right)}{NetSellVol_{i,G,d}}.$$

Notice the $BuyVolCNY_{i,d,j}$ and $SellVolCNY_{i,d,j}$ in the numerator are cash volumes, products of trading share volumes and trading prices, rather than share volumes. Variable $ret_{i,d+1,d+n}$ and $rf_{d+1,d+n}$ are the cumulative return of stock *i* and the risk-free rate from day $d+1$ to day $d+n$, respectively. Trading cost is computed as follows:

$$TrdCost_{d,G} = \sum_{i}\sum_{j\in G} \tau_1 \times BuyVolCNY_{i,d,j} + \sum_{i}\sum_{j\in G} \tau_2 \times SellVolCNY_{i,d,j}. \tag{A7}$$

Variables τ_1 and τ_2 capture the proportional trading costs relative to cash volumes. According to China's policies and practices, there are commissions of 0.05% of cash volumes on both the buy and sell sides, a stamp tax of 0.10% on cash volumes of sales, and a transfer fee of 0.002% on cash volumes on both the buy and sell sides during our sample period. In other words, $\tau_1 = 0.05\% + 0.002\%$, $\tau_2 = 0.05\% + 0.10\% + 0.002\%$.

Three additional considerations are added in the third step. First, BLLO assumes that the net buy and sell portfolios are held for *n* days and sets *n* to different horizons. As the main discussion of BLLO focuses on the holding horizon of 140 days (*n*=140), we choose the same horizon for ease of comparison. Second, BLLO assumes that the net buy (sell) portfolios are independent for each day *d*, and annualizes the performance by aggregating the daily net buy and sell portfolios over each year.[2] Third, to have a perspective of total performance as a percentage of the total investment of these tracking portfolios, which is similar to the idea of return on investments, BLLO measures total investment as the aggregate holding value for each group of investors. For comparison, we choose the aggregate holding values in Panel A of Table I and present the return on investment as the total performance over aggregate holding.

[2] We compute the standard error estimates using the Newey and West (1987) method, with the optimal bandwidth chosen by using Newey and West (1994) procedure. We conduct robustness check using optimal lag numbers selected by the Bayesian Information Criterion (BIC). The results are similar and available on request.

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After computing the total performance, BLLO decomposes the total into three intuitive components,

$$Total_{d,G} = StkSelect_{d,G} + MktTiming_{d,G} - TrdCost_{d,G}. \quad \text{(A8)}$$

Here, the stock selection component on day *d* for investor group *G* is computed as:

$$StkSelect_{d,G} = \sum_{i \in NetBuy_{d,G}} NetBuyVol_{i,G,d} \times AvgBuyPrice_{i,G,d} \times (ret_{i,d+1,d+n} - rmkt_{d+1,d+n}) - \sum_{i \in NetSell_{d,G}} NetSellVol_{i,G,d} \times AvgSellprice_{i,G,d} \times (ret_{i,d+1,d+n} - rmkt_{mkt,d+1,d+n}). \quad \text{(A9)}$$

In other words, BLLO compares each stock's return, $ret_{i,d+1,d+n}$, with the contemporaneous market return, $rmkt_{d+1,d+n}$, to determine whether investors are capable of choosing stocks that beat the market. The second component, the market timing component, is computed as:

$$MktTiming_{d,G} = \sum_{i \in NetBuy_{d,G}} NetBuyVol_{i,G,d} \times AvgBuyPrice_{i,G,d} \times (rmkt_{d+1,d+n} - rf_{d+1,d+n}) - \sum_{i \in NetSell_{d,G}} NetSellVol_{i,G,d} \times AvgSellprice_{i,G,d} \times (rmkt_{d+1,d+n} - rf_{d+1,d+n}). \quad \text{(A10)}$$

With the difference between the market portfolio and interest rate, the BLLO method aims to capture whether the investor can choose the allocation between stocks (market portfolio) and bonds (interest rate), and thus, time the market. The trading cost term is defined in equation (A7).

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Appendix Table I. Previous Studies on Retail Investors in Different Markets

Publication Details	Data	Research Questions and Main Findings
Barber and Odean, Journal of Finance, 2000. "Trading is hazardous to your wealth: The common stock investment performance of individual investors?"	U.S. data, 66,465 households over 1991 to 1996 from a large discount brokerage	Research Question: Return performance of equities held directly by households. Findings: (1) Individual investors who hold common stocks directly pay a tremendous performance penalty for active trading. (2) Overconfidence can explain high trading levels and the resulting poor performance of individual investors.
Kaniel, Saar, and Titman, Journal of Finance, 2008. "Individual investor trading and stock returns."	U.S. data, all retail orders from NYSE CAUD file from year 2000 to 2003	Research Question: The relation between net individual investor trading and short-horizon cross-sectional stock returns. Findings: (1) Individuals buy stocks following declines in the previous month and sell following price increases. (2) Positive excess returns in the month following intense buying by individuals and negative excess returns after individuals sell.
Barber, Odean, and Zhu, Review of Financial Studies, 2008. "Do retail trades move markets?"	U.S. data, tick-by-tick trades from TAQ and ISSM over 1983 to 2001, small trades proxy for retail	Research Question: The trading of individual investors and stock returns. Findings: (1) Small trade order imbalance correlates well with order imbalance based on trades from retail brokers. (2) Retail investors herd. (3) Small trade order imbalance negatively forecasts future annual returns. (4) Small trade order imbalance positively predicts future week returns.
Kaniel, Liu, Saar, and Titman, Journal of Finance, 2012. "Individual investor trading and return patterns around earnings announcements."	U.S. data, all retail orders from NYSE CAUD file from year 2000 to 2003	Research Question: The informed trading by individual investors around earnings announcements Findings: (1) The intense aggregate individual investor buying (selling) predicts large positive (negative) abnormal returns on and after earnings announcement dates. (2) Decompose abnormal returns following the event into information and liquidity provision components, and show that about half of the returns can be attributed to private information.

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Publication Details	Data	Research Questions and Main Findings
Kelley and Tetlock, Journal of Finance, 2013. “How wise are crowds? Insights from retail orders and stock returns.”	U.S. data, retail orders from dozens of retail brokerages over year 2003 to 2007	Research question: The role of retail investors in stock pricing by separately examining aggressive (market) and passive (limit) orders. Findings: (1) Both market and limit order imbalance positively predict firms’ monthly stock returns. (2) Market orders correctly predict firm news, including earnings surprises. (3) Limit orders following negative returns, consistent with traders providing liquidity.
Boehmer, Jones, Zhang and Zhang, Journal of Finance, 2021. “Tracking retail investor activity.”	U.S. data, an algorithm to identify marketable retail trades from TAQ over year 2010 to 2015	Research Question: Provide an algorithm to identify marketable retail purchases and sales using TAQ. Findings: (1) Provide and validate the algorithm. (2) Marketable retail order imbalance positive predict future weekly stock return. (3) Predictive power of marketable retail order imbalance could be attributable to order flow persistence, contrarian trading, public news sentiment and unexplained part.
Barber, Lin, and Odean, Journal of Financial and Quantitative Analysis, forthcoming . “Resolving a paradox: Retail trades positively predict returns but are not profitable.”	U.S. data, retail trades identified by Boehmer et al. (2021) from TAQ over year 2010-2019	Research Question: Retail order imbalance positively predicts returns, but in aggregate retail investor trades lose money. Findings: (1) Order imbalance tests equally weight stocks, but retail purchases concentrate in stocks that subsequently underperform. (2) Trades by retail investors with less knowledge, experience, and wealth are more likely to underperform.
Welch, Journal of Finance, 2022. “The wisdom of the Robinhood crowd.”	U.S. data, Robinhood investors holding from May 2018 to August 2020	Research Question: Robinhood investors trading behavior. Findings: (1) Robinhood investors increased their holdings in the March 2020 COVID bear market. (2) Robinhood investors tend to buy stocks with high past share volume and dollar-trading volume. (3) From mid-2018 to mid-2020, an aggregated Robinhood portfolio had both good timing and good alpha.

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Publication Details	Data	Research Questions and Main Findings
Ozik, Sadka, and Shen, Journal of Financial and Quantitative Analysis, 2021. "Flattening the illiquidity curve: Retail trading during the COVID-19 lockdown."	U.S. data, Robinhood investors holding from May 2018 to August 2020	Research Question: The impact of retail investors on stock liquidity during the Coronavirus pandemic lockdown. Findings: (1) Retail trading exhibits a sharp increase during Covid. (2) Retail trading attenuated the rise in illiquidity by roughly 40%, but less so for high-media-attention stocks.
Barber, Huang, Odean, and Schwarz, Journal of Finance, 2022. "Attention-Induced Trading and Returns: Evidence from Robinhood Users."	U.S. data, Robinhood investors holding from May 2018 to August 2020	Research Question: The influence of financial innovation by fintech brokerages on individual investors' trading and stock prices. Findings: (1) Robinhood investors engage in more attention-induced trading than other retail investors. (2) Intense buying by Robinhood users forecasts negative returns. Average 20-day abnormal returns are −4.7% for the top stocks purchased each day.
Grinblatt and Keloharju, Journal of Financial Economics, 2000. "The investment behavior and performance of various investor types: a study of Finland's unique data set."	Finnish data, daily retail trades and holdings from Finnish Central Securities Depository (FCSD) over 1994 to 1996	Research Question: The past-return-based behavior and the performance of various investor types. Findings: (1) Foreign investors tend to be momentum, while domestic investors, particularly households, tend to be contrarians. (2) The portfolios of foreign investors seem to outperform the portfolios of households, even after controlling for behavior differences.
Linnainmaa, Journal of Finance, 2011. "Do limit orders alter inferences about investor performance and behavior?"	Finnish data, daily retail trades and holdings from Finnish Central Securities Depository (FCSD) over 1995 to 2002	Research Question: Individual investors' trading behaviors could be explained by investors' use of limit orders. Findings: These patterns arise because limit orders are price-contingent and suffer from adverse selection.

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Publication Details	Data	Research Questions and Main Findings
Grinblatt, Keloharju, Linnainmaa, Journal of Financial Economics, 2012. “IQ, trading behavior, and performance.”	Finnish data, daily retail trades and holdings from Finnish Central Securities Depository (FCSD) over 1995 to 2002 and intelligence (IQ) test administered to Finnish male	Research Question: Whether IQ influences trading behavior, performance, and transaction costs. Findings: (1) High-IQ investors are less subject to disposition effect, more aggressive about tax-loss trading, and more likely to supply liquidity when stock experience a one-month high. (2) High-IQ investors exhibit superior market timing, stock-picking skill, and trade execution.
Bach, Calvet and Sodini, American Economic Review, 2020. “Rich pickings? Risk, return, and skill in household wealth.”	Sweden data, annual administrative panel that reports the full balance sheet of Swedish residents between 2000 and 2007	Research Question: Examine the risk and return characteristics of household wealth. Findings: (1) The expected return on household net wealth increases with net worth. (2) The expected wealth return is driven by systematic risk-taking and exhibits strong persistence. Idiosyncratic risk is transitory but sufficiently large to generate substantial long-term dispersion in returns. (3) Wealth returns explain most of the historical increase in top wealth shares.
Dorn, Huberman and Sengmueller, Journal of Finance, 2008. “Correlated trading and returns.”	German data, 37,000 retail clients at one of the three largest German discount brokers from 1998 to 2000	Research Question: Investors correlated trading and stock returns. Findings: (1) Investors tend to be on the same side of the market. (2) Correlated market orders lead returns due to persistent speculative price pressure. (3) Correlated limit orders also predict subsequent returns, consistent with liquidity demands.
Barrot, Kaniel, and Sraer, Journal of Financial Economics, 2016. “Are retail traders compensated for providing liquidity?”	French data, from a leading European online broker between 2002 and 2010	Research Question: Whether individual investors provide liquidity to the stock market and whether they are compensated for doing so. Findings: (1) The ability of aggregate retail order imbalances to predict short-term future returns is significantly enhanced during times of market stress. (2) Individual investors do not reap the rewards from liquidity provision because they experience a negative return on the trading day and reverse their trades for too long after the liquidity provision dissipated.

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Publication Details	Data	Research Questions and Main Findings
Fong, Gallagher, and Lee, Journal of Financial and Quantitative Analysis, 2014. "Individual investors and broker types"	Australian data, transaction data from Australian Securities Exchange (ASX) SIRCA with identification of the broker between 1995 and 2007	Research Question: Examine the informativeness of trades via discount and full-service retail brokers. Findings: (1) Trades via full-service retail brokers are more informative than are trades via discount retail brokers. (2) Past returns, volatility, and news announcements could explain the net volume of discount retail brokers, but could not explain the net volume of full-service retail brokers.
Barber, Lee, Liu, and Odean, Review of Financial Studies, 2009. "Just how much do individual investors lose by trading?"	Chinese Taiwan data, transaction data in the Taiwan stock exchange between 1995 and 1999	Research Question: How much do individual investors lose by trading? Findings: (1) The aggregate portfolio of individuals suffers an annual performance penalty of 3.8%. (2) Nearly all individual trading losses can be traced to their aggressive orders.
Barber, Lee, Liu, and Odean, Journal of Financial Markets, 2014. "The cross-section of speculator skill: Evidence from day trading"	Chinese Taiwan data, day traders' transaction data in the Taiwan stock exchange between 1992 and 2006	Research Question: Examine the cross-sectional differences of returns earned by speculative day traders. Findings: Less than 1% of the day trader population can predictably and reliably earn positive abnormal returns net of fees.
Balasubramaniam, Campbell, Ramadorai and Ranish, Journal of Finance, 2023. "Who owns what? A factor model for direct stock holding"	Indian data, 10 million retail investors holdings data on August 2011from Indian stock market	Research Question: Build a cross-sectional factor model for retail investors' direct stock holdings. Findings: (1) Stock characteristics such as firm age and share price have strong investor clienteles. (2) Account attributes such as account age, account size, and extreme under diversification are associated with particular characteristic preferences. (2) Coheld stocks have higher return covariance.

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Publication Details	Data	Research Questions and Main Findings
Anagol, Balasubramaniam, and Ramadorai, Journal of Financial Economics, 2021. “Learning from noise: Evidence from India’s IPO lotteries”	Indian data, 1.5 million investors participate in allocation lotteries for 54 IPO stocks between 2007 and 2012	Research Question: Retail investors participate in allocation lotteries for Indian IPO stocks. Findings: Investors who wind the IPO lottery and obtain IPO stocks that rise in value increase portfolio trading volume in non-IPO stocks relative to lottery losers. A learning model could explain the results.
Titman, Wei, and Zhao, Journal of Financial Economics, 2022. “Corporate actions and the manipulation of retail investors in China: An analysis of stock splits”	Chinese data, account trading data from a major stock exchange in China between 1999 and 2015	Research Question: Corporate actions and the manipulation of retail investors in the stock splits events. Findings: (1) Share prices temporarily increase after stock splits, and subsequently decline below their presplit levels. (2) Small retail investors buy shares in firms initiating suspicious splits, while more sophisticated investors buy before suspicious split announcements and sell in the postsplit period. (3) Insiders sell large blocks of shares and obtain loans using company stock as collateral before the suspicious splits.
An, Lou, and Shi, Journal of Monetary Economics, 2022. “Wealth redistribution in bubbles and crashes.”	Chinese data, account trading from a major stock exchange in China, 2014-15 bubbles and crashes episode	Research Question: What are the social-economic consequences of financial market bubbles and crashes? Findings: The largest 0.5% households in the equity market gain, while the bottom 85% lose, 250B RMB through active trading in this period.
Liao, Peng, Zhu, Review of Financial Studies, 2021. “Extrapolative bubbles and trading volume.”	Chinese data, account-level transaction data from one of the largest brokerage firms between 2014 and 2015	Research Questions: Propose an extrapolative model to explain the sharp rise in prices *and* volume during financial bubbles. Findings: (1) The model proposes a novel mechanism for volume: because of extrapolative beliefs and disposition effects, investors are quick to buy assets with positive past returns and sell them if good returns continue. (2) Use Chinese account-level data to confirm the model’s predictions.

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Publication Details	Data	Research Questions and Main Findings
Chen, Gao, He, Jiang, and Xiong, Journal of Econometrics, 2019. “Daily price limits and destructive market behavior.”	**Chinese data**, account trading from a major stock exchange in China from between 2012 and 2015	**Research Question:** Daily price limits and investors trading behavior. **Findings**: Large investors tend to buy on the day when a stock hits the 10% upper price limit and then sell on the next day; and their net buying on the limit-hitting day predicts stronger long-run price reversal.
Li, Geng, Subrahmanyam and Yu, Journal of Empirical Finance, 2017. “Do wealthy investors have an informational advantage? Evidence based on account classifications of individual investors”	**Chinese data,** a brokerage firm providing one million investors’ trading records between January 2007 and 2009	**Research Question:** Do wealthy investors have an informational advantage? **Findings:** (1) Wealthy investors with portfolio values above the 99.5th percentile (“super” investors) outperform all other investors. (2) Part of their excess returns could be explained by informational advantages.
Liu, Peng, Xiong, and Xiong, Journal of Financial Economics, 2022. “Taming the bias zoo.”	**Chinese data,** Combine subjective survey responses in 2018 with account-level transaction data from a major stock exchange in China	**Research Question:** Many biases offer observationally similar predictions for a targeted financial anomaly. This study combines subjective survey responses with observational data to tame the bias zoo. **Findings:** In cross-sectional regressions of respondents’ actual turnover on survey-based trading motives, perceived information advantage, and gambling preference dominate other motives**.**

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Appendix Table II. Comparing Main Findings of This Study with the Results from Previous Literature

Our main findings	**Findings from previous literature**
Return Predictability: Retail investors with smaller account sizes negatively predict future returns, whereas retail investors with larger account sizes positively predict future returns.	Retail trading negatively predicts future returns: Barber and Odean (2000), Grinblatt and Keloharju (2000), Barber, Huang, Odean, and Schwarz (2022) Retail trading positively predicts future returns: Kaniel, Saar, and Titman (2008), Barber, Odean, and Zhu (2008), Dorn, Huberman and Sengmueller (2008), Grinblatt, Keloharju, Linnainmaa (2012), Kelley and Tetlock (2013), Fong, Gallagher, and Lee (2014), Barber, Lee, Liu, and Odean (2014), Barrot, Kaniel, and Sraer (2016), Li, Geng, Subrahmanyam, and Yu (2017), Boehmer, Jones, Zhang, and Zhang (2021), Welch (2022)
Momentum/Contrarian: Retail investors with smaller account sizes display daily momentum and weekly contrarian patterns, whereas retail investors with larger account sizes display contrarian patterns.	Retail investors display momentum trading patterns: Kelley and Tetlock (2013), Boehmer, Jones, Zhang, and Zhang (2021) Retail investors display contrarian trading patterns: Grinblatt and Keloharju (2000), Kaniel, Saar, and Titman (2008), Linnainmaa (2011), Grinblatt, Keloharju, Linnainmaa (2012), Barrot, Kaniel, and Sraer (2016), Welch (2022)
Process Information: Retail investors with smaller account sizes fail to process public news, whereas retail investors with larger account sizes incorporate public news in trading.	Retail investors can process information: Kaniel, Liu, Saar, and Titman (2012), Kelley and Tetlock (2013), Boehmer, Jones, Zhang, and Zhang (2021) Retail investors fail to process information: Li, Geng, Subrahmanyam, and Yu (2017), Titman, Wei, and Zhao (2022)
Behavior Biases: Retail investors with smaller account sizes show behavioral patterns such as overconfidence and gambling preferences, while retail investors with larger account sizes could better predict returns in stocks more attractive to investors with behavioral biases.	Retail investors display behavioral biases: Barber and Odean (2000), Liu, Peng, Xiong, and Xiong (2022), Barber, Huang, Odean, and Schwarz (2022) Some retail investors display behavioral biases, while others trade against behavioral biases: Chen, Gao, He, Jiang, and Xiong (2019)

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Appendix Table III. Distributions of Investor Trading and Holding

This table reports the summary statistics for the trading and holding of different investor groups across different stock characteristics. Our sample period covers January 2016 to June 2019. Our sample firms are A-share stocks listed on a major stock exchange with at least 15 days of trade during the previous month. The trading and holding for different investor groups across different stock characteristics are reported in Panels A and B, which are the time-series averages of cross-sectional means. The bottom two and top two sectors, in terms of trading volume and holding within each investor group, are reported in Panels C and D. The sectors are classified according to the Datastream Level 4 sector classification. Panel E reports the trading volumes of different investor groups across different trade sizes, which are the time-series averages of cross-sectional means.

Panel A. Trading volumes across different stock characteristics

	RT1	RT2	RT3	RT4	RT5	INST	CORP
Small	6.3%	23.3%	32.5%	13.4%	14.0%	9.9%	0.7%
Medium	5.8%	21.2%	31.3%	13.3%	13.9%	13.7%	0.8%
Large	4.3%	16.3%	26.7%	12.9%	15.7%	22.4%	1.8%
Low EP	6.3%	22.0%	31.4%	13.7%	15.3%	10.5%	0.8%
Medium EP	5.6%	21.2%	30.7%	12.9%	13.8%	14.9%	0.9%
High EP	4.3%	17.1%	28.0%	13.1%	14.7%	21.2%	1.6%
Low Turnover	4.7%	17.5%	28.0%	12.9%	14.5%	20.7%	1.7%
Medium Turnover	5.2%	19.4%	29.9%	13.5%	15.0%	15.8%	1.1%
High Turnover	6.3%	23.4%	32.1%	13.2%	14.2%	10.1%	0.6%

Panel B. Holding shares across different stock characteristics

	RT1	RT2	RT3	RT4	RT5	INST	CORP
Small	4.1%	13.4%	20.2%	9.9%	17.4%	7.1%	27.9%
Medium	3.6%	10.8%	16.5%	8.0%	15.4%	11.1%	34.6%
Large	2.2%	6.6%	10.6%	5.3%	10.1%	16.7%	48.6%
Low EP	3.9%	11.2%	16.9%	8.5%	15.5%	8.2%	35.7%
Medium EP	3.4%	11.3%	17.3%	8.2%	15.6%	12.6%	31.6%
High EP	2.4%	7.3%	12.0%	6.2%	11.8%	14.5%	45.8%
Low Turnover	2.2%	5.8%	9.3%	4.9%	11.2%	11.9%	54.6%
Medium Turnover	3.1%	9.0%	14.7%	7.8%	15.5%	12.9%	37.0%
High Turnover	4.3%	15.0%	22.2%	10.1%	16.2%	10.5%	21.6%

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Panel C. Sectors with lowest and highest trading volumes

	RT1	RT2	RT3	RT4	RT5	INST	CORP
Bottom1	**Banks & Life Insurance** 2.4%	**Banks & Life Insurance** 10.5%	**Banks & Life Insurance** 20.4%	**Banks & Life Insurance** 11.9%	**Alternative Energy** 12.2%	**Alternative Energy** 6.5%	**Alternative Energy** 0.4%
Bottom2	Financial Services 3.6%	Financial Services 15.8%	Beverages 27.4%	Health Care Equipment & Services 11.9%	Gas, Water and Multiutilities 12.3%	Industrial Metals & Mining 11.6%	General Industrials 0.7%
Top2	Industrial Metals & Mining 6.5%	Industrial Engineering 22.3%	Gas, Water and Multiutilities 32.0%	Technology Hardware & Equipment 14.5%	Software & Computer Services 17.7%	Travel & Leisure 21.3%	Financial Services 2.9%
Top1	**Alternative Energy** 7.6%	**Alternative Energy** 25.5%	**Alternative Energy** 34.2%	**Financial Services** 14.5%	**Banks & Life Insurance** 18.0%	**Banks & Life Insurance** 32.8%	**Banks & Life Insurance** 4.1%

Panel D. Sectors with lowest and highest holding shares

	RT1	RT2	RT3	RT4	RT5	INST	CORP
Bottom1	**Banks & Life Insurance** 1.1%	**Banks & Life Insurance** 4.0%	**Banks & Life Insurance** 7.6%	**Banks & Life Insurance** 4.2%	**Banks & Life Insurance** 5.6%	**Alternative Energy** 2.9%	**Household Goods & Home Construction** 19.2%
Bottom2	Aerospace & Defense 1.9%	Aerospace & Defense 7.2%	Food & Drug Retailers 12.1%	Industrial Transportation 5.6%	Electricity 7.3%	Mobile Telecommunications 4.8%	Health Care Equipment & Services 23.2%
Top2	Forestry & Paper 4.5%	Household Goods & Home Construction 12.8%	Support Services 19.2%	Support Services 9.7%	Support Services 19.5%	Banks & Life Insurance 23.3%	Banks & Life Insurance 54.3%
Top1	**Alternative Energy** 6.1%	**Alternative Energy** 15.5%	**Alternative Energy** 20.9%	**Forestry & Paper** 9.8%	**Software & Computer Services** 23.8%	**Health Care Equipment & Services** 23.5%	**Mobile Telecommunications** 55.1%

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Panel E. Trading volumes across different trade sizes

Trade Size	RT1	RT2	RT3	RT4	RT5	INST	CORP
<40,000 CNY	4.78%	10.86%	7.44%	1.22%	0.74%	5.35%	0.10%
40,000-200,000 CNY	0.69%	8.47%	14.54%	5.04%	3.65%	4.36%	0.27%
>200,000 CNY	0.00%	1.03%	8.19%	6.96%	10.30%	5.48%	0.73%

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Appendix Table IV. Long-short Portfolios

This table reports an alternative method for Tables III and IV using long-short portfolios. On each day, we first group stocks into different counterparty bins in Panel A and into different days-to-cover ratios in Panel B. Within each bin, we then sort stocks into three groups according to the order imbalances of different investor groups. We form a long-short portfolio by longing the stock with the highest 1/3 order imbalance measures and shorting those with the lowest 1/3 order imbalance measures. The long-short portfolios are held on the next day and the portfolio returns are adjusted by Liu, Stambaugh and Yuan (2019) three-factor model. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Daily alphas for long-short portfolios with different counterparties

RT1-4	RT5	INST	Sample Coverage		RT1-RT4	RT5	INST
Buy	Buy	Sell	17%	Alpha	-0.0004**	0.0003**	0.0019***
				[*t*-stat]	[-2.16]	[2.32]	[11.05]
Buy	Sell	Buy	11%	Alpha	-0.0006**	0.0002	-0.0003*
				[*t*-stat]	[-2.33]	[0.83]	[-1.85]
Buy	Sell	Sell	21%	Alpha	-0.0005***	0.0011***	0.0020***
				[*t*-stat]	[-2.75]	[8.36]	[11.30]
Sell	Buy	Buy	22%	Alpha	-0.0028***	0.0011***	0.0009***
				[*t*-stat]	[-14.81]	[6.74]	[6.11]
Sell	Buy	Sell	13%	Alpha	-0.0033***	0.0023***	-0.0004**
				[*t*-stat]	[-9.10]	[10.88]	[-1.99]
Sell	Sell	Buy	17%	Alpha	-0.0011***	-0.0003*	0.0007***
				[*t*-stat]	[-6.60]	[-1.86]	[4.33]

Panel B. Daily alphas for long-short portfolios with different counterparties

Holding Horizon		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5	INST
(0,10]Days	Alpha	-0.0054***	-0.0048***	-0.0034***	-0.0003*	0.0039***	0.0027***
	[*t*-stat]	[-19.22]	[-17.82]	[-14.72]	[-1.67]	[12.40]	[13.30]
(10,20]Days	Alpha	-0.0028***	-0.0023***	-0.0016***	-0.0006***	0.0014***	0.0021***
	[*t*-stat]	[-14.54]	[-11.45]	[-8.39]	[-4.63]	[7.80]	[9.85]
(20,60]Days	Alpha	-0.0020***	-0.0014***	-0.0011***	-0.0003***	0.0006***	0.0019***
	[*t*-stat]	[-14.70]	[-8.19]	[-7.25]	[-4.05]	[5.79]	[14.08]
Above60Days	Alpha	-0.0025***	-0.0049***	-0.0056***	-0.0017**	0.0003***	0.0015***
	[*t*-stat]	[-4.56]	[-5.47]	[-4.49]	[-2.52]	[3.07]	[12.84]

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Appendix Table V. Predict Returns over the Next 12 Weeks Using Previous Week Order Imbalances

This table reports the alternative results for Table V using previous week order imbalances to predict the cross-section of returns over the next 12 weeks. The independent variables are the previous week's order imbalance, *Oib*(*w*-1). The control variables are the same as those in Table II, and are not reported for brevity. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

*w*th weeks		OibRT1(*w*-1)	OibRT2(*w*-1)	OibRT3(*w*-1)	OibRT4(*w*-1)	OibRT5(*w*-1)	OibINST(*w*-1)
1	Estimate	-0.0449***	-0.0428***	-0.0280***	-0.0041***	0.0064***	0.0081***
	[*t*-stat]	[-12.37]	[-11.26]	[-8.70]	[-3.42]	[8.05]	[12.14]
2	Estimate	-0.0176***	-0.0162***	-0.0095***	0.0004	0.0034***	0.0031***
	[*t*-stat]	[-7.64]	[-7.18]	[-5.09]	[0.39]	[5.17]	[5.15]
3	Estimate	-0.0099***	-0.0082***	-0.0050**	-0.0003	0.0031***	0.0021***
	[*t*-stat]	[-3.83]	[-2.93]	[-2.05]	[-0.33]	[5.64]	[2.93]
4	Estimate	-0.0072***	-0.0065***	-0.0036*	-0.0002	0.0023***	0.0008
	[*t*-stat]	[-3.48]	[-2.93]	[-1.71]	[-0.21]	[3.54]	[1.42]
5	Estimate	-0.0059***	-0.0053***	-0.0050***	-0.0025**	0.0009	0.0011**
	[*t*-stat]	[-3.60]	[-2.93]	[-2.71]	[-2.58]	[1.57]	[2.14]
6	Estimate	-0.0098***	-0.0089***	-0.0062***	-0.0010	0.0003	0.0021***
	[*t*-stat]	[-4.73]	[-4.24]	[-3.42]	[-1.04]	[0.65]	[4.14]
7	Estimate	-0.0070***	-0.0060***	-0.0035**	-0.0004	0.0009	0.0018***
	[*t*-stat]	[-4.11]	[-3.44]	[-2.21]	[-0.33]	[1.53]	[4.46]
8	Estimate	-0.0048**	-0.0033	-0.0027	-0.0002	0.0013**	0.0010*
	[*t*-stat]	[-2.49]	[-1.53]	[-1.39]	[-0.16]	[2.02]	[1.80]
9	Estimate	-0.0030	-0.0015	-0.0011	0.0006	0.0010*	0.0002
	[*t*-stat]	[-1.46]	[-0.75]	[-0.62]	[0.80]	[1.81]	[0.39]
10	Estimate	-0.0035*	-0.0030	-0.0024	0.0001	-0.0007	0.0010*
	[*t*-stat]	[-1.66]	[-1.38]	[-1.22]	[0.11]	[-1.25]	[1.72]
11	Estimate	-0.0011	-0.0007	0.0009	0.0008	0.0001	0.0001
	[*t*-stat]	[-0.54]	[-0.29]	[0.41]	[0.77]	[0.19]	[0.10]
12	Estimate	-0.0027	-0.0027	-0.0004	-0.0005	0.0000	0.0001
	[*t*-stat]	[-1.22]	[-1.20]	[-0.19]	[-0.63]	[0.05]	[0.20]

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Appendix Table VI. Two Stage Decomposition with Additional Lags of Order Imbalances

This table presents the alternative two-stage decomposition results for Table VI with additional lags of order imbalances. Panel A reports the first-stage estimation results controlling additional two weeks order imbalance. The order imbalances are decomposed into five components as specified in the following equation:

$$Oib_{i,d} = p0_d + \left[p1_d^1 Oib_{i,d-1} + p1_d^2 Oib_{i,d-2,d-6} + p1_d^3 Oib_{i,d-7,d-11}\right] + p2'_d Ret_{i,d-1} + p3_d Overconf_{i,d-1} + p4_d Gamble_{i,d-1} + u10_{i,d}, \quad \text{(A11)}$$

where the order persistence components contain additional two weeks order imbalances, and other components are same as in Table VI. Panel B reports the second-stage decomposition of the order imbalance's predictive power, as specified in the following equation:

$$Oib_{i,d} = \widehat{Oib}_{i,d}^{persistence^{MoreLags}} + \widehat{Oib}_{i,d}^{liquidity} + \widehat{Oib}_{i,d}^{overconf} + \widehat{Oib}_{i,d}^{gamble} + \widehat{Oib}_{i,d}^{other}, \quad \text{(A12)}$$

where the order imbalance component, *Oib*(-1,*Persistence*^{MoreLags}), is constructed using the previous one day and previous two weeks order imbalance in first stage, and the rest are the same as those in Table VI. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. First stage of projecting the order imbalance on persistence, past returns, overconfidence, and gambling proxies

		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib(-1)	Estimate	0.1517***	0.1675***	0.1496***	0.0456***	0.0917***
	[*t*-stat]	[24.43]	[36.07]	[37.47]	[15.01]	[28.14]
Oib(-6,-2)	Estimate	0.2434***	0.2246***	0.1931***	0.0950***	0.1524***
	[*t*-stat]	[38.71]	[37.27]	[33.78]	[18.14]	[30.75]
Oib(-11,-7)	Estimate	0.0691***	0.0541***	0.0533***	0.0475***	0.0691***
	[*t*-stat]	[12.43]	[10.33]	[11.53]	[13.47]	[22.22]
Ret(-1)	Estimate	0.5303***	0.6989***	0.4174***	-0.2229***	-1.3117***
	[*t*-stat]	[9.16]	[15.30]	[13.53]	[-7.67]	[-30.37]
Ret(-6,-2)	Estimate	-0.2261***	-0.0271**	0.0435***	-0.0285	-0.0799***
	[*t*-stat]	[-12.25]	[-2.54]	[3.85]	[-1.60]	[-4.95]
Ret(-27,-7)	Estimate	0.0077*	0.0074**	-0.0033	-0.0282***	-0.0170***
	[*t*-stat]	[1.95]	[2.28]	[-1.19]	[-7.51]	[-3.99]
Overconf(-1)	Estimate	0.0391***	0.0204**	0.0218***	0.0167	-0.0533***
	[*t*-stat]	[3.11]	[2.17]	[2.90]	[1.63]	[-5.05]
Gamble(-1)	Estimate	0.0172	0.0580***	0.1430***	0.2229***	-0.0425*
	[*t*-stat]	[1.03]	[4.49]	[11.54]	[12.02]	[-1.94]

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Intercept	Estimate	-0.0131***	-0.0076***	-0.0080***	-0.0062***	0.0174***
	[*t*-stat]	[-4.14]	[-3.60]	[-6.05]	[-3.75]	[8.57]
Adj.R2		9.68%	7.64%	5.37%	1.21%	2.99%

Panel B. Second stage decomposition of order imbalance's predictive power

Dep.var		Ret	Ret	Ret	Ret	Ret
Oib.var		OibRT1	OibRT2	OibRT3	OibRT4	OibRT5
Oib(-1,Persistence^{MoreLags})	Estimate	-0.0284***	-0.0274***	-0.0202***	-0.0062***	0.0059***
	[*t*-stat]	[-13.95]	[-13.52]	[-11.19]	[-3.95]	[8.36]
Oib(-1,Liquidity)	Estimate	-0.0085**	-0.0174***	-0.0255***	0.0204	0.0037
	[*t*-stat]	[-2.16]	[-3.87]	[-3.03]	[1.29]	[1.50]
Oib(-1,Overconf)	Estimate	-0.1022	-0.4609***	-0.6067***	-0.9331***	0.1815***
	[*t*-stat]	[-1.63]	[-3.79]	[-4.29]	[-5.06]	[5.14]
Oib(-1,Gamble)	Estimate	-0.5580***	-0.1310***	-0.0538***	-0.0453***	0.3008***
	[*t*-stat]	[-4.87]	[-4.00]	[-4.05]	[-5.32]	[6.35]
Oib(-1,Other)	Estimate	-0.0084***	-0.0082***	-0.0058***	-0.0008***	0.0011***
	[*t*-stat]	[-15.19]	[-15.17]	[-13.90]	[-6.06]	[9.40]
Adj.R2		10.54%	10.37%	10.06%	9.61%	9.48%
Interquartile return						
Oib(-1,Persistence^{MoreLags})		-0.1387%	-0.1103%	-0.0656%	-0.0123%	0.0335%
Oib(-1,Liquidity)		-0.0124%	-0.0223%	-0.0195%	0.0102%	0.0089%
Oib(-1,Overconf)		-0.0150%	-0.0338%	-0.0404%	-0.0428%	0.0252%
Oib(-1,Gamble)		-0.0317%	-0.0251%	-0.0254%	-0.0334%	0.0422%
Oib(-1,Other)		-0.1694%	-0.1433%	-0.0980%	-0.0227%	0.0475%

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Appendix Table VII. Annual Performance Using Tracking Portfolios of Retail Trading, from Liquidity Provision and Liquidity Demand

This table reports the annual performance for different retail investor groups using the similar framework as in Table VIII and from the liquidity provision and demand perspective. The annual total performance could be decomposed into liquidity provision, liquidity demand, and trading cost, as specified in the following equation:

$$Total_{d,G} = LiquidityProvision_{d,G} + LiquidityDemand_{d,G} - TrdCost_{d,G}. \tag{A13}$$

We assume that if investors trade direction is opposite to direction of previous stock price movement, then investors provide liquidity; and if investors trade in the same direction of previous stock price movement, then investor demand liquidity. The total performance from liquidity provision and liquidity demand are defined in the following equations:

$$\begin{aligned} &LiquidityProvision_{d,G} \\ &= \sum\nolimits_{i \in NetBuy_{d,G}\ and\ ret_{i,d-1} \leq 0} NetBuyVol_{i,G,d} \times AvgBuyPrice_{i,G,d} \times (ret_{i,d+1,d+n} - rf_{d+1,d+n}) \\ &- \sum\nolimits_{i \in NetSell_{d,G}\ and\ ret_{i,d-1} \geq 0} NetSellVol_{i,G,d} \times AvgSellPrice_{i,G,d} \times (ret_{i,d+1,d+n} - rf_{d+1,d+n}). \end{aligned} \tag{A14}$$

$$\begin{aligned} &LiquidityDemand_{d,G} \\ &= \sum\nolimits_{i \in NetBuy_{d,G}\ and\ ret_{i,d-1} > 0} NetBuyVol_{i,G,d} \times AvgBuyPrice_{i,G,d} \times (ret_{i,d+1,d+n} - rf_{d+1,d+n}) \\ &- \sum\nolimits_{i \in NetSell_{d,G}\ and\ ret_{i,d-1} < 0} NetSellVol_{i,G,d} \times AvgSellPrice_{i,G,d} \times (ret_{i,d+1,d+n} - rf_{d+1,d+n}). \end{aligned} \tag{A15}$$

Panel A presents the total annual trading performance by different investor groups, from liquidity provision and demand perspective. Panel B further decompose the annual trading performance into stock selection and market timing components. Panel C reports the annual trading performance around earnings announcements. The earnings announcement events include the stock on announcement day *d* and the previous day *d*-1. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

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Panel A. Annual trading performance using tracking portfolios

		Total	Liquidity Provision	Liquidity Demand	Trading Cost
RT1	Performance	-5.61%***	-0.34%	-3.89%***	-1.38%***
	[*t*-stat]	[-10.29]	[-0.82]	[-7.65]	[-23.47]
RT2	Performance	-4.61%***	-0.04%	-2.78%***	-1.80%***
	[*t*-stat]	[-8.96]	[-0.09]	[-6.45]	[23.07]
RT3	Performance	-3.60%***	-0.25%	-1.63%***	-1.72%***
	[*t*-stat]	[-9.49]	[-0.90]	[-6.06]	[-20.89]
RT4	Performance	-2.80%***	-0.11%	-1.11%***	-1.58%***
	[*t*-stat]	[-8.51]	[-0.49]	[-4.63]	[-19.53]
RT5	Performance	-0.29%	1.89%***	-1.14%***	-1.03%***
	[*t*-stat]	[-0.92]	[5.12]	[-3.05]	[-18.84]
INST	Performance	1.15%***	0.84%***	0.73%***	-0.41%***
	[*t*-stat]	[3.57]	[5.70]	[2.91]	[-20.49]

Panel B. Annual trading performance using tracking portfolios: separate stock selection and market timing

		Total	Liquidity Provision		Liquidity Demand		TradingCost
			Stock Selection	Market Timing	Stock Selection	Market Timing	
RT1	Performance	-5.61%***	0.07%	-0.41%	-4.10%***	0.22%	-1.38%***
	[*t*-stat]	[-10.29]	[0.25]	[-1.39]	[-8.51]	[1.03]	[-23.47]
RT2	Performance	-4.61%***	0.43%	-0.46%*	-2.97%***	0.19%	-1.80%***
	[*t*-stat]	[-8.96]	[1.46]	[-1.79]	[-7.63]	[1.12]	[23.07]
RT3	Performance	-3.60%***	0.17%	-0.42%**	-1.77%***	0.14%	-1.72%***
	[*t*-stat]	[-9.49]	[0.85]	[-2.47]	[-7.48]	[1.02]	[-20.89]
RT4	Performance	-2.80%***	0.33%**	-0.44%***	-1.23%***	0.12%	-1.58%***
	[*t*-stat]	[-8.51]	[2.01]	[-3.32]	[-6.66]	[0.65]	[-19.53]
RT5	Performance	-0.29%	2.45%***	-0.56%***	-1.40%***	0.26%	-1.03%***
	[*t*-stat]	[-0.92]	[7.96]	[-3.03]	[-5.32]	[1.18]	[-18.84]
INST	Performance	1.15%***	0.89%***	-0.05%	0.49%***	0.23%	-0.41%***
	[*t*-stat]	[3.57]	[6.71]	[-0.62]	[3.04]	[1.32]	[-20.49]

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Panel C. Annual trading performance using tracking portfolios: around earnings announcements

		Total		Liquidity Provision		Liquidity Demand		Trading Cost	
		EA	EA/all days	EA	EA/all days	EA	EA/all days	EA	EA/all days
RT1	Performance	-0.39%***	6.89%	-0.05%	14.43%	-0.30%***	7.60%	-0.04%***	3.06%
	[*t*-stat]	[-3.68]		[-0.86]		[-4.10]		[-5.91]	
RT2	Performance	-0.36%***	7.71%	-0.03%	93.81%	-0.27%***	9.54%	-0.06%***	3.12%
	[*t*-stat]	[-4.08]		[-0.64]		[-3.89]		[-5.90]	
RT3	Performance	-0.24%***	6.61%	-0.03%	11.05%	-0.16%***	9.52%	-0.05%***	3.18%
	[*t*-stat]	[-4.24]		[-0.75]		[-3.44]		[-5.86]	
RT4	Performance	-0.14%***	5.02%	0.00%	0.24%	-0.09%**	8.01%	-0.05%***	3.26%
	[*t*-stat]	[-2.70]		[-0.01]		[-2.06]		[-5.86]	
RT5	Performance	0.10%	-33.28%	0.19%***	9.90%	-0.06%	4.96%	-0.03%***	3.37%
	[*t*-stat]	[0.91]		[3.13]		[-1.18]		[-5.87]	
INST	Performance	0.11%*	9.80%	0.10%***	11.57%	0.03%	4.23%	-0.01%***	3.56%
	[*t*-stat]	[1.81]		[3.48]		[0.67]		[-5.86]	

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Appendix Table VIII. Gender and Age

This table examines retail investors by gender and age. The sample period covers January 2019 to March 2019. The sample firms are A-share stocks with at least 15 days of trade during the previous month. Panel A reports the summary statistics of trading volumes across age and gender groups. Panel B reports the return predictions across genders and age groups. To account for serial correlation in the coefficients, the standard errors of the time series are adjusted using Newey and West (1987) method, with optimal bandwidths chosen by following Newey and West (1994). ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Summary Statistics of gender and age groups

Gender	Trading Volume (% of total)	
Age	<45	>=45
Male	29%	38%
Female	13%	20%

Panel B. Cross-sectional return predictions, by different gender and age groups

Dep.var		Ret	Ret	Ret	Ret
Gender		Male	Male	Female	Female
Age		<45	>45	<45	>45
Oib(-1)	Estimate	-0.0026***	-0.0060***	0.0007	-0.0002
	[*t*-stat]	[-5.45]	[-3.48]	[1.46]	[-0.32]
	Interquartile return	-0.05%	-0.11%	0.02%	0.00%

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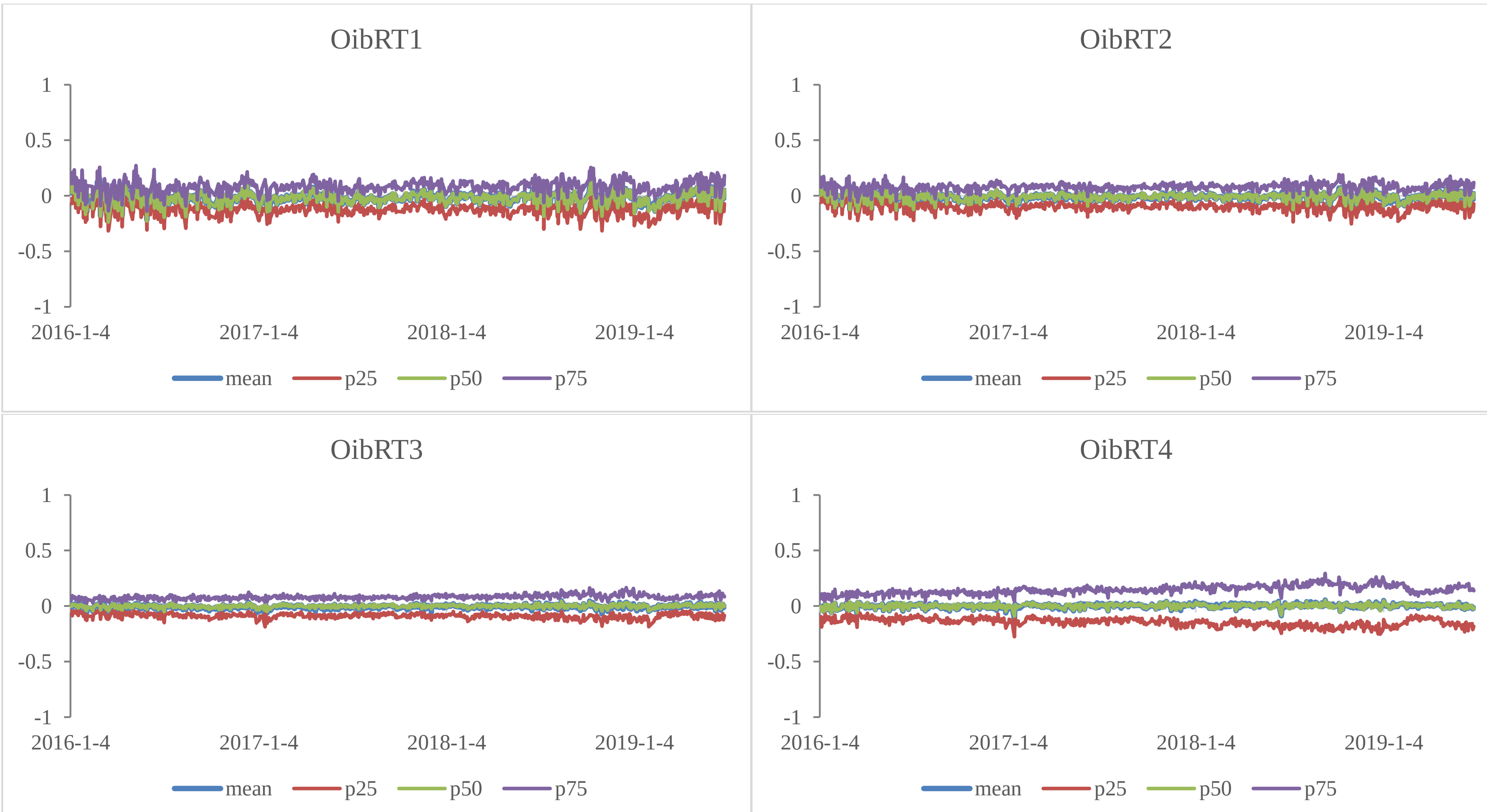
Appendix Figure I. Time Series of Different Types of Investor Order Imbalance

These figures report the time series of the different types of investor trading activities. The sample period covers January 2019 to March 2019. The sample firms are A-share stocks with at least 15 days of trade during the previous month. We present the cross-sectional mean, median, 25th percentiles, and 75th percentiles of scaled daily order imbalances for each investor group each day. Order imbalance is computed as the buy share volume minus the sell share volume divided by the buy share volume plus sell share volume for each investor group, as specified in Equation (1).



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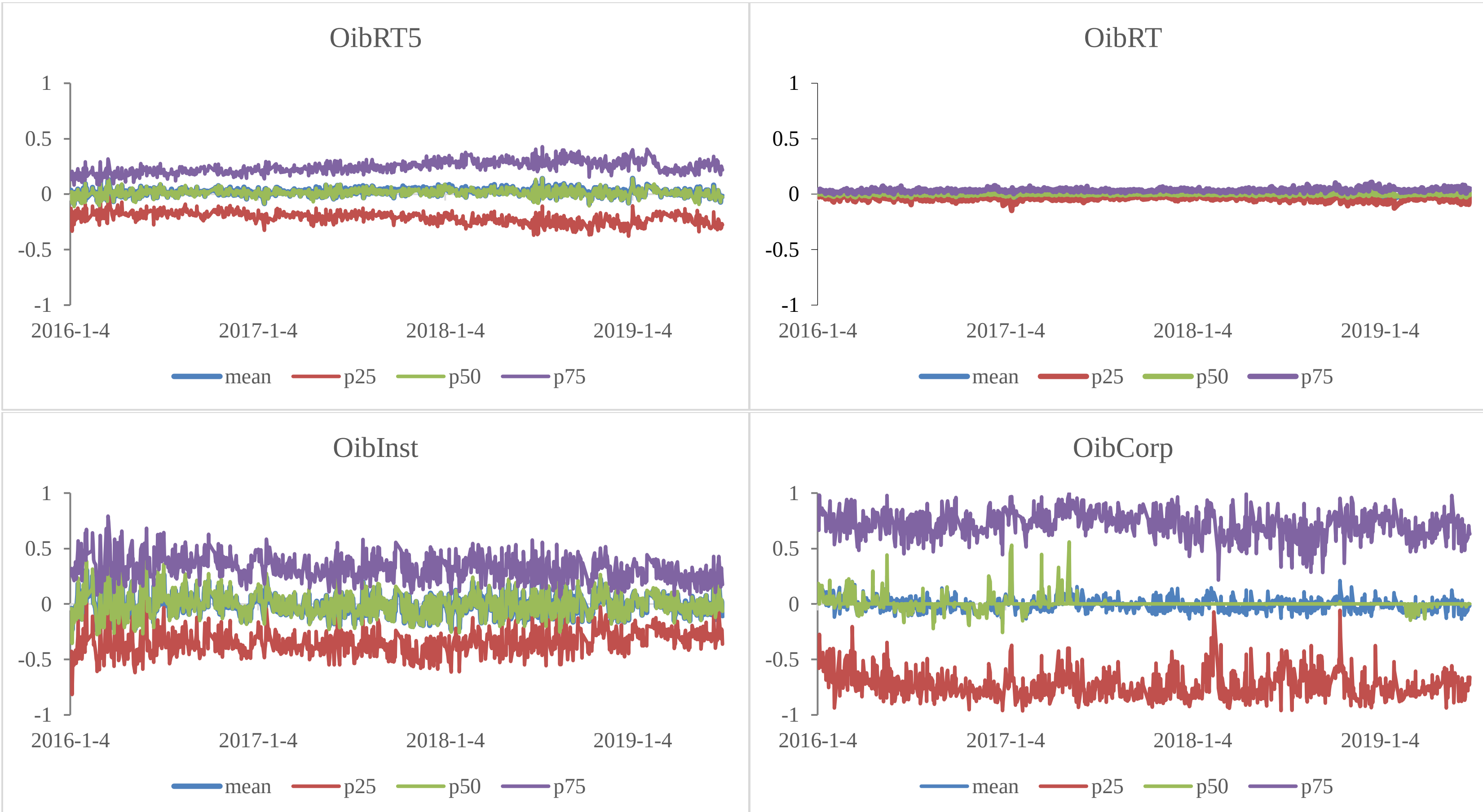

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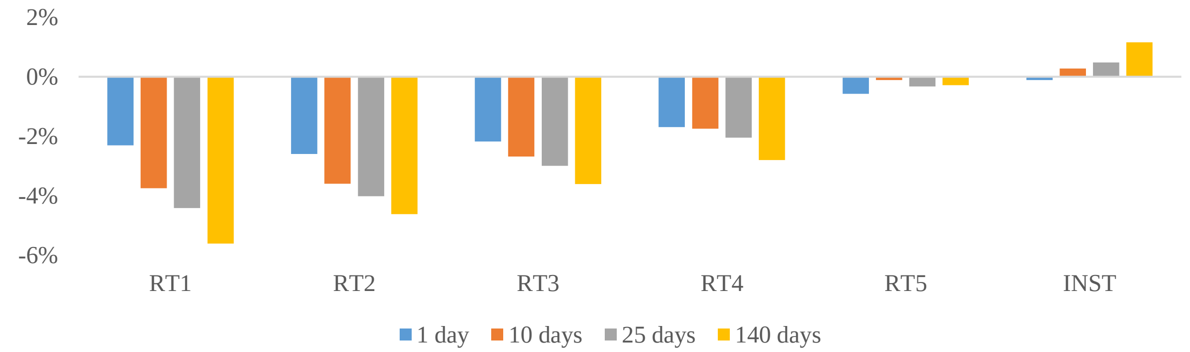
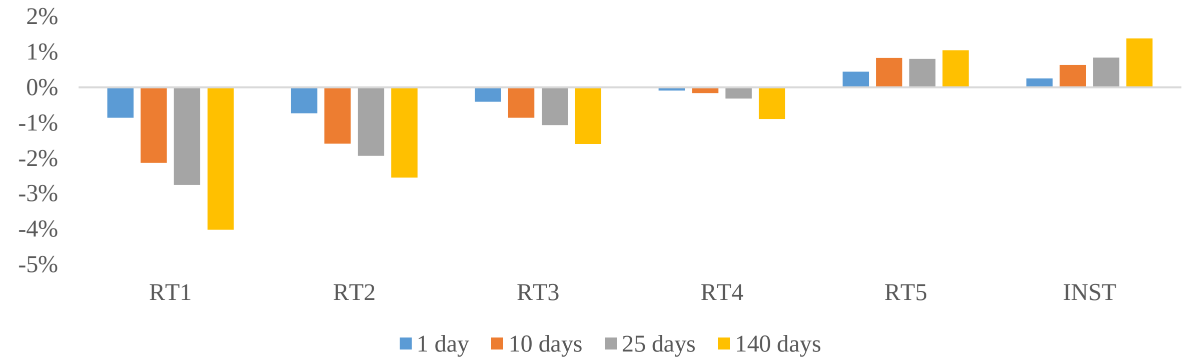
Appendix Figure II. Annual Performances over Different Portfolio Holding Horizon

This table reports the alternative performance for different retail investor groups over different portfolio holding days. Following Barber et al. (2009) method, as specified in equation (A4) to (A10), we construct the portfolios tracking aggregate retail trading and present the annual performance for holding n days (n = 1 day, 10 days, 25 days, and 140 days). Panel A presents the annual total performance and Panel B and C present stock selection and market timing.

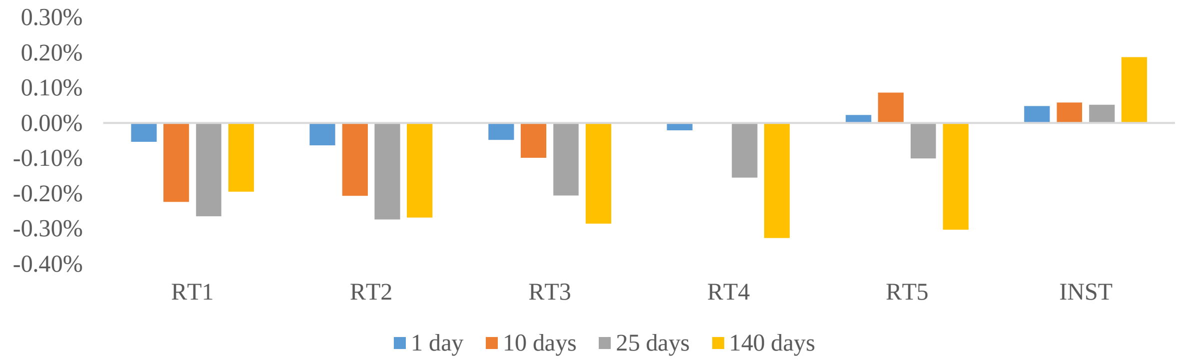
Panel A. Total trading performances over different portfolio holding days



Panel B. Stock selection over different portfolio holding days



Panel C. Market timing over different portfolio holding days



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